

Multi-step stability selects sparse surrogate models for economic greenhouse climate control: an in silico study of feature-library design and actuator-pathway survival

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Abstract: Economic model predictive control (MPC) can coordinate crop-climate requirements and resource costs in greenhouse production, but comparing candidate process models at crop scale is expensive. We therefore conducted an in silico study using the published GreenLight tomato greenhouse model driven by ERA5 weather. Seventy-two sparse-identification configurations were screened, and fifteen controllers were compared over four 60-day test seasons with up to 20 identification replicates. Outcomes were simulated seasonal margin and climate-corridor violations. One-step accuracy and multi-step stability ranked the feature libraries oppositely: physically motivated features reduced one-step temperature RMSE from 1.86 to 1.68 °C but increased median 24 h rollout RMSE from 2.67 to 24.27 °C. The library selected by multi-step stability achieved the highest mean margin (+4.32 EUR m⁻²), compared with +3.09 for the best physics-informed variant and +2.26 for a tuned agronomic heuristic. Conditioning did not explain the non-monotone margin pattern (+4.32, +0.28, +2.75); performance instead tracked survival of the direct boiler coefficient after sparsification (55, 15, 55 %). Five controllers remained on the margin–violation Pareto front. These results support rollout-based, actuator-aware surrogate screening for greenhouse MPC, but establish comparative simulation evidence rather than field performance.

Keywords: greenhouse climate control; protected tomato production; economic model predictive control; sparse identification of nonlinear dynamics; simulation-based evaluation; control-oriented model selection; multi-step prediction error; actuator-pathway survival

1. Introduction

Protected tomato production depends on maintaining a productive microclimate while containing the variable costs of heating, supplementary lighting and CO₂ enrichment. These inputs affect crop growth and resource use simultaneously, so a control strategy should be judged agronomically and economically rather than by setpoint tracking alone [1–2]. Economic model predictive control (MPC) provides that framework: inputs are optimised over a finite horizon subject to temperature, humidity and CO₂ constraints [3–6].

Simulation-based greenhouse studies are established in this application area. Xu et al. used a greenhouse–crop model to compare open- and closed-loop optimal control and to estimate lettuce yield and profit in an explicitly computational study [7]. Wen et al. combined measured blueberry photosynthesis with an SVR model and simulated multi-objective light–CO₂ regulation [8], while Ećim-urić et al. coupled a TRNSYS greenhouse model to an artificial neural network and anchored the thermal model to measured temperatures [9]. These examples show both the value and the evidential limit of simulation: virtual experiments can isolate controller and model-design effects, but field performance requires empirical anchoring. The present study is deliberately at the simulation-only end of that spectrum. It uses the independently published GreenLight tomato model and recorded weather forcing, but introduces no new greenhouse, plant or actuator measurements. Its claims are therefore comparative statements inside a defined virtual testbed, not demonstrations of commercial greenhouse performance.

MPC requires a process model, and calibrating a detailed mechanistic greenhouse model for every site is laborious. A surrogate identified from simulated or measured trajectories can reduce that burden. Sparse identification of nonlinear dynamics (SINDy) is attractive because it expresses the dynamics as a sparse combination of library functions [10–13]. Unlike a neural surrogate, the resulting equations can be inspected for physical and dimensional consistency and embedded analytically in an MPC solver.

Three benchmarking risks complicate an objective comparison of such surrogates. First, learned controllers may be compared with an unsearched heuristic even though tuned agronomic rules are a strong reference [14–16]. Second, a setpoint-tracking index may be reported instead of economic margin [6,14], although the two need not be monotonically related. Third, surrogate hyperparameters are commonly selected by one-step prediction accuracy [12,17], implicitly assuming that the best short-horizon predictor also gives the best controller. Table 1 states the safeguards used here and quantifies sensitivity to removing them.

The third assumption is precisely what identification-for-control research warns against: a model should be judged by the loop it closes, not by open-loop fit alone [18–20]. In model-based reinforcement learning the same problem is termed *objective mismatch*: predictive accuracy can be a poor proxy for control performance [21–23], and recent surveys show that it remains relevant for modern data-driven surrogates [24–25]. Exhaustive closed-loop screening is expensive even in simulation because it requires a solver and a full cropping season per candidate. The practical question is therefore narrower: if a greenhouse surrogate must be selected open loop, which inexpensive criterion best anticipates seasonal closed-loop economics?

A related question concerns feature-library design. Physically motivated basis functions such as saturation vapour pressure, vapour-pressure deficit and screen-attenuated radiation are often included on the reasonable assumption that known physics can only help [26–27]. Yet these functions are nonlinear transforms of the same climate states and may be nearly collinear with them; collinearity can destabilise sparse regression [28–29]. We therefore test, rather than assume, whether physically motivated features improve a control-oriented surrogate.

Table 1. Three benchmarking risks, the safeguards used in this study and the measured sensitivity to removing each safeguard. All monetary values are simulated seasonal margin in EUR m^{−2} unless stated otherwise.

Benchmarking risk	Safeguard used here	Observed sensitivity
Unsearched reference: learned controllers	The agronomic rule-based reference is tuned on the training seasons only, by a 16-trial random	Tuning lifts the heuristic from −1.23 to +2.26 (gain +3.49), so every advantage measured against the untuned reference is overstated by

Benchmarking risk	Safeguard used here	Observed sensitivity
compared with a hard-coded heuristic	search — the same trial budget the reinforcement-learning agents receive	that amount. The tuned heuristic buys the gain with constraint pressure (2813 to 4339 violation steps) and is itself off the Pareto front
Objective mismatch: tracking error reported instead of margin, or MPC actuator weights that are inconsistent with the prices used for evaluation	Every controller is scored by the simulator’s own profit function, jointly with constraint violations under Pareto dominance; the MPC stage cost is re-weighted to match the price vector	Re-weighting improves both axes for all eight MPC controllers: +1.54 to +2.74 for the five physics-library variants and +1.57 for the neural surrogate, against +0.21 to +0.25 for the two raw-library ones, cutting the apparent library effect from +3.66 to +1.23
Open-loop selection: identification configuration chosen by one-step prediction accuracy	Candidates are scored by multi-step rollout stability and regression conditioning, fixed before and independently of any closed-loop evaluation	Among the candidates that are embeddable in the solver, the best one-step configuration comes from the library with condition number 53.4 rather than the one with 8.2, and with median 24-h rollout RMSE 24.27 °C rather than 2.67 °C

We address both questions in a controlled in silico experiment using the published GreenLight tomato model [1,16], ERA5-derived weather for one location and a 15-min control step. Candidate surrogates were identified on two training seasons, and fifteen controllers were evaluated — nine SINDy-MPC variants, a neural-network MPC, two deep reinforcement-learning agents [30–32], a full-simulator-model MPC and the rule-based heuristic in both its stock and tuned configurations — over four held-out test seasons. A *season* here is the 60-day window from 1 March, and the outcome is model-derived margin per square metre of greenhouse floor area. Each learned controller was run at 20 identification replicates per season (10 for the neural-network MPC); the heuristic is deterministic and was run once per season, and the full-simulator-model MPC returned only three of the four seasons, its 2022 runs having exhausted the solver-failure budget. Comparisons use non-parametric signed-rank tests with Holm correction [33–35]; against the deterministic heuristic the test is one-sample against a constant rather than genuinely paired. The contributions are four.

(i) One-step accuracy and multi-step stability rank candidate libraries in opposite orders, and the closed-loop ordering follows the multi-step one. The identification ladder evaluated 72 labelled configurations, covering 54 distinct fits — two of the optimizer labels turn out to be bit-identical. Within its degree-1 block — the degree-2 candidates are all non-embeddable in the MPC solver — the library with the best one-step accuracy is the worst on every multi-step metric. Taking the undenoised cells under the two sparse estimators ($n = 40$ fits per library), adding physically motivated features improves one-step RMSE from 1.86 to 1.68 °C while degrading the median 24-h rollout RMSE from 2.67 to 24.27 °C, raising the regression condition number from 8.2 to 53.4 and the divergence rate from 0.000 to 0.076 (Figure 1); withholding only the bilinear cross terms recovers part of the loss but not the ordering (rollout 10.58 °C, $\kappa = 24.5$). In closed loop, a raw, physics-free library — also the sparsest of the three — attains the best four-season mean margin of the fifteen controllers, +4.32 EUR m⁻² against +3.09 for the best physics-informed variant, ahead in 62 of 80 season $\times$ replicate pairs ($p_{\text{Holm}} = 4.6 \times 10^{-4}$, family of 15). The gap is small against the run-to-run spread — standard deviations 4.08 and 2.71 for a difference of 1.23 — and it reverses in one of the four seasons (−1.56 against +0.27 in 2021). The library ordering holds again on four seasons never used at any stage of model or controller development (four controllers, eight replicates, default objective). Within this virtual testbed, physical motivation alone is therefore not sufficient evidence that a basis function belongs in the control-oriented library.

(ii) The effect of physically motivated features is non-monotone. All three libraries were carried into closed loop under a strict one-factor design — same estimator, same sparsity threshold, same seeds, same priced objective, four seasons $\times$ 20 replicates, zero truncated runs, the library the only thing that changes. Open-loop degradation is monotone in the number of physics features: $\kappa = 8.2 \rightarrow 24.5 \rightarrow 53.4$ and median rollout $2.67 \rightarrow 10.58 \rightarrow 24.27$ °C. Closed-loop margin is not: $+4.32 \rightarrow +0.28 \rightarrow +2.75$ EUR m^{-2} . The worst-conditioned library beats the middle one by $+2.47$ on average (median $+2.39$, 63 of 80 paired runs, $p = 1.3 \times 10^{-8}$), in each of the four seasons taken separately. Under the second sparse estimator only the two endpoints exist as closed-loop controllers ($+3.83$ raw, $+2.48$ full physics), so the raw-over-physics direction replicates there but the three-point non-monotonicity rests on one estimator. Conditioning therefore predicts open-loop multi-step stability cleanly but is not a sufficient explanation of closed-loop economics; contribution (iv) identifies the controller-facing quantity that better tracks the outcome.

(iii) Benchmarking safeguards materially change the apparent effect size. Against the tuned heuristic the raw-library controller is better by $+2.06$ EUR m^{-2} , ahead in 75 of 80 season $\times$ replicate comparisons ($p_{\text{Holm}} = 6.3 \times 10^{-11}$, Holm correction over a family of 15). Three further SINDy-MPC variants also clear the tuned heuristic significantly, by smaller margins ($+1.57$, $+0.83$, $+0.82$), and the two full-physics controllers are ahead of it in point estimate without being separable from it: the raw library is first, not alone. Against the unsearched reference the same advantage would read $+5.54$, demonstrating the sensitivity of the headline comparison to baseline tuning. Aligning actuator weights with the evaluation price vector likewise reduces the raw-minus-physics gap from $+3.66$ to $+1.23$ (Table 1, row 2).

(iv) What the closed-loop outcome does track is survival of the actuator pathway through the sparsity threshold. Across the one-factor triple, the direct boiler coefficient $\xi_{u\text{Boil}}$ is retained in 11 of 20 identification replicates with the raw library, 11 of 20 with full physics and 3 of 20 with physics-without-cross-terms — 55 %, 55 %, 15 %, which is the margin ordering and not the κ ordering. A surrogate whose boiler column has been thresholded away cannot be steered with the boiler, however accurate it is one step ahead and however well conditioned the fit that produced it. The controlled evidence is a single-coefficient ablation, the only randomised manipulation in the mechanism block: reinstating the deleted coefficient in the frozen recipe, with the library, the scalars, every other term and the MPC construction held fixed, is positive in 16 of 20 replicates ($p_{\text{Holm}} = 1.2 \times 10^{-3}$ over the two knock tests) with median $+0.21$ EUR m^{-2} and mean $+1.92$ — significant, typically small and heavily right-skewed — while the complementary knock-out is null. We are explicit about the limits of this account. Survival *gates* the outcome but does not *order* it: among the two libraries that keep the term the raw one still wins by $+1.56$ EUR m^{-2} , and raw-library runs that lost the term still average $+3.48$, above physics-library runs that kept it ($+2.36$). The obvious structural account of why the middle library is the worst — that it carries enough collinearity for the threshold to destabilise the direct boiler term but, unlike full physics, has no bilinear $T_{\text{in}}u_{\text{Boil}}$ term to carry the heating pathway once the direct one is cut — has been tested and rejected: deleting exactly that one feature from the full library leaves conditioning ($\kappa = 52.3$), open-loop instability and closed-loop margin ($+2.11$ EUR m^{-2} , $p_{\text{Holm}} = 4.4 \times 10^{-6}$ above the middle library) at the full library's level, with survival at 40 % — nowhere near the collapse a bilinear bypass predicts (Sections 3.5 and 4). Why one library loses the term is thus unresolved; that survival gates the outcome is not.

What this does not show is equally explicit. The advantage is not Pareto-dominant: the front holds five points, and the PPO agent attains far fewer constraint violations (1330 against 4130 steps per season) at a lower margin. The surrogate carries no crop state, so the MPC objective is necessarily a proxy and terminal biomass cannot be priced. The

mechanism experiments are single-season and replication is unequal across controllers. The non-monotonicity rests on the middle library, which reaches closed loop under one estimator only, and survival is measured rather than manipulated at library level — the ablation is the only randomised contrast anywhere in the mechanism, and the term-deletion test that removed the detour explanation is, like the library comparison itself, a controlled structural comparison rather than a randomised one. All results are simulation-only, for one location and one planting date, and reproducibility is established within one computing environment only. Consequently, the reported margins and rankings are comparative computational evidence, not estimates of field-scale benefit or recommendations for direct commercial deployment. The remainder of the paper is organised as follows. Section 2 describes the controlled object, the economic criterion, the data, the controllers and the identification protocol; Section 3 reports the model-selection ladder, the four-season economic comparison and the mechanism experiments; Section 4 discusses the implications and the limits; Section 5 concludes.

2. Materials and Methods

2.1. Simulation study design, site, weather and cropping season

This study is an in silico comparative experiment: no new greenhouse sensor records, plant measurements, harvested-yield observations or actuator trials were collected. All controller development and evaluation used the published GreenLight tomato greenhouse model [1–2] through the GreenLight-Gym interface [16]. GreenLight provides a common mechanistic plant–climate testbed, so controller and surrogate choices can be changed while the virtual crop and greenhouse are held fixed. It does not remove model-form error or the simulation-to-reality gap; those limits constrain the interpretation in Section 4.6.

The simulated house is at Rostov-on-Don (47.24° N, 39.71° E). Weather forcing came from the ERA5/ERA5-Land reanalysis [36] via Open-Meteo [37]; sky temperature was derived from the retrieved dew point and cloud cover, and the deep-soil boundary condition was refitted as a single annual harmonic to ERA5-Land soil temperature at this site, replacing the simulator’s Dutch default. Outdoor CO₂ was held at 400 ppm, a limitation of the model. Each season spans 60 days from 1 March at a 15-min control step ($T_s = 900$ s), i.e. 5760 control steps per season; that count is the truncation gate used throughout.

Years were separated to prevent leakage: identification and every selection decision used 2018–2019 only, closed-loop evaluation used 2020–2023 (2020 in-distribution, 2021–2023 out-of-distribution), and a further block of four seasons, 2014–2017, was generated after all recipes had been frozen and entered no stage of training, selection or tuning. Training trajectories were collected by the rule-based controller with a pseudo-random binary excitation of amplitude 0.3 on its actions plus an additive perturbation of scale 0.1; the condition number of the regression matrix was recorded for every fit [28]. Evaluation rollouts carry no excitation.

2.2. Economic criterion

Controllers are scored by the profit term of the simulator’s own reward function; no tracking-error proxy is used. At step k that term is a marginal income π_k , and the seasonal economic performance indicator (EPI), written J throughout, is

$$J = \sum_k \pi_k = \sum_k (g_k - c_k) \quad (1),$$

where g_k is income from modelled fruit dry-matter growth, converted to fresh mass with a dry-to-fresh ratio of 0.065 and valued at 1.60 EUR/kg, and c_k are the variable costs of heating (0.09 EUR/(kW · h)), supplementary lighting (0.30 EUR/(kW · h)) and CO₂

enrichment (0.30 EUR/kg). Dividing the logged cost columns by the logged energy and CO₂ columns of regen/results/final/main.csv returns exactly these three prices across all 780 retained runs. J is *not* the scalar the environment returns as its reward: that scalar rescales the profit and subtracts normalised corridor-violation penalties. The distinction matters, because the two reinforcement-learning controllers are trained on the penalised reward while every controller is scored on the unpenalised J — their position on the violation axis partly reflects optimising a different objective from the one reported.

What J excludes matters for every result that follows. It omits planting material, labour, water, fertiliser, depreciation and fixed costs; it uses modelled dry-matter growth rather than realised marketable yield; and, critically, it does not value the biomass standing in the house at the end of the 60-day window. The surrogates of Section 2.4 carry a three-component climate state and no crop state, so terminal biomass cannot be priced inside the optimisation even in principle. That is why the MPC stage cost (Section 2.6) is a proxy rather than a discretisation of Equation (1), and why J is reported as a *margin* and not as a farm profit: a controller that harvests within the window is rewarded over one that builds a crop for a later window, and no result here separates the two.

J carries no penalty for leaving the productive range, so margin can be bought with violations. The simulator’s productive band is CO₂ ∈ [300,1600] ppm, air temperature [15,34] °C and relative humidity [50,85] %. Controllers are therefore compared by Pareto dominance in two axes — J and the seasonal count of steps in violation — and no scalarised “margin-minus-penalty” criterion is introduced, because the simulator does not express violations in money. This corridor is also not the one the predictive controllers minimise against: their stage cost (Section 2.6) penalises humidity only *above* 85 % and CO₂ only *below* a floor, so the lower humidity and upper CO₂ bounds are unweighted in the optimisation yet counted on the violation axis.

2.3. Controllers compared

Fifteen controllers were compared (Table 2). The *low-threshold* SINDy variant is not a reduced first-principles grey box; it is the same sparse estimator on the same feature library at a threshold of 10^{−6}, i.e. the least sparse member of the family. Second, the aggregation loop applied to two SINDy controllers rolls the learner’s own MPC in closed loop on the training seasons, appends the visited states to the identification set and re-estimates the model (3 iterations of 5-day episodes). The rule-based controller is *never queried* in that loop: this is on-policy re-identification, not DAgger and not imitation learning, and the rule-based controller is not an expert in it. One asymmetry inside that loop survives into the headline comparison and is disclosed here: its episodes are rolled under the default stage cost, so under the priced objective these two controllers aggregate their training states under one cost function and are evaluated under another.

Table 2. Controllers compared. Identification recipes are read from the frozen configuration regen/regen_config.py; “library” refers to Section 2.4. All predictive controllers share one horizon ($N = 20$) and one solver-failure budget (100).

Label	Family	Definition
rule_based	heuristic	agronomic setpoint heuristic, hard-coded setpoints; deterministic
rule_based_tuned	heuristic	the same heuristic after a 16-trial setpoint search (Section 2.7)
sindy_mpc_raw	SINDy-MPC	raw library, STLSQ, $\lambda = 0.05$
sindy_mpc_raw_ens	SINDy-MPC	raw library, ensemble, $\lambda = 0.05$
sindy_mpc_conf	SINDy-MPC	frozen recipe: physics_no_cross, ensemble, $\lambda = 0.05$

Label	Family	Definition
sindy_mpc_phys	SINDy-MPC	physics (full, with cross terms), STLSQ, $\lambda = 0.05$
sindy_mpc_phys_ens	SINDy-MPC	physics (full, with cross terms), ensemble, $\lambda = 0.05$
sindy_mpc_dense	SINDy-MPC	physics_no_cross, STLSQ, $\lambda = 10^{-3}$
sindy_mpc_lowthr	SINDy-MPC	physics_no_cross, STLSQ, $\lambda = 10^{-6}$ (not a grey box)
sindy_mpc_conf_dagger	SINDy-MPC	conf plus on-policy re-identification
sindy_mpc_dense_dagger	SINDy-MPC	dense plus on-policy re-identification
nn_mpc	neural MPC	64×64 MLP surrogate on the physics library, 300 epochs; same horizon and stage cost, but the network is not embeddable in the NLP, so its MPC is a shooting formulation solved by L-BFGS-B on autograd gradients (25 iterations per step)
ppo, sac	reinforcement learning	PPO [30] and SAC [31] from Stable-Baselines3 [32], $2 \cdot 10^5$ environment steps
oracle_mpc	idealised	cross-entropy planning directly on the simulator integrator: 48 candidate sequences, 2 refinement iterations, elite fraction 0.2

Four of the SINDy-MPC labels form a deliberate one-factor design over the feature library, and the mechanism analysis of Section 3.5 rests on it. Holding the polynomial degree (1), the denoiser (none), the estimator and the sparsity threshold (0.05) at the frozen recipe's values, and changing *only* θ , gives `sindy_mpc_raw_ens` (raw), `sindy_mpc_conf` (physics_no_cross) and `sindy_mpc_phys_ens` (physics) under the ensemble estimator, and `sindy_mpc_raw` and `sindy_mpc_phys` under STLSQ. The ensemble arm is a complete three-point series; the STLSQ arm has only the two endpoints, because the two physics_no_cross STLSQ controllers were frozen at thresholds 10^{-3} and 10^{-6} rather than 0.05. Any statement about how the library ordering behaves *between* the endpoints therefore rests on the ensemble arm alone, and is scoped that way throughout.

The 16-trial budget given to the rule-based reference (Section 2.7) matches the recorded budget of the reinforcement-learning controllers, so the reference is not an unsearched baseline. That match is a stated budget, not a measured search trace: here the agents are trained once per seed at stock Stable-Baselines3 defaults and the neural surrogate at a fixed architecture, and no hyper-parameter search artefact for either exists in the results tree. The prediction horizon was *not* tuned for any predictive controller either; that asymmetry runs against the MPC family and is disclosed in the Discussion.

2.4. Feature libraries and sparse identification

Sparse regression [10–12] recovers an explicit discrete one-step map

$$x_{k+1} = \Xi \Theta(x_k, u_k, d_k) \quad (2),$$

with state $x_k = (T_{\text{in}}, c, \varphi)^T$ (air temperature, CO₂, relative humidity), six controls u_k (boiler, CO₂ injection, thermal screen, ventilation, lamps, blackout screen) normalised to [0,1], and disturbances d_k (outdoor temperature, global radiation, outdoor CO₂, two diurnal harmonics). Three feature libraries were compared, differing only in θ :

- raw — 11 features: the five disturbance channels and the six controls, with no state transform whatsoever;
- physics_no_cross — those 11 plus three physically motivated transforms of state, radiation and screen position (14 features): saturation vapour pressure $e_s(T_{\text{in}}) = 0.6108 \exp(17.27 T_{\text{in}} / (T_{\text{in}} + 237.3))$ by the Magnus–Tetens relation, vapour-pressure deficit $D = (1 - \varphi/100) e_s$, and screen-attenuated radiation $S_{\text{eff}} = I (1 - u_{\text{thscr}})$;

- physics — those 14 plus four bilinear state $\times$ control cross terms ($T_{in}S_{eff}$, φu_{vent} , $(c - c_{out})u_{vent}$, $T_{in}u_{boil}$), 18 features. 286 287

The three physical functions are nonlinear transforms of variables already in the raw library: they add no new measurement, only near-collinear columns. States and features are standardised before thresholding, so the threshold acts on dimensionless quantities, and the coefficients are transformed back to physical units. Four estimators were available: STLSQ, SR3, a constrained variant, and an ensemble optimiser with 20 bagged bootstrap replicates [13]. Because the ensemble re-draws its bootstrap, a single fit is one realisation of a lottery; the draw is therefore carried as an explicit design axis, and the number of draws is recorded per row rather than assumed — 10 per seed in the canonical wave, 6 per seed in the later wave that added the raw-library controller. 288 289 290 291 292 293 294 295 296

2.5. Identification ladder and pre-registered open-loop criteria 297

The identification recipe was selected in an open-loop ladder computed on the training years alone: 3 libraries $\times$ 2 polynomial degrees $\times$ 4 estimators $\times$ 3 denoisers = 72 configuration labels, each fitted on 20 seeds, for 1440 fits (regen/results/ladder_rerun/). No closed-loop quantity enters it. Each fit records the one-step RMSE, the multi-step rollout RMSE and divergence fraction at horizons of 4, 20 and 96 steps (1 h, 5 h, 24 h; the reported value is always the 24-h figure), the number of non-zero terms, the regression condition number, and whether the map can be embedded analytically in the MPC. The pre-registration declared two gates, MPC-embeddability and a sign-and-dimension transparency check, and specified multi-step stability only qualitatively, as a divergence fraction near zero. What was applied differs on both counts and is reported as applied: the divergence criterion was operationalised as a fixed threshold of 0.05 in the table generator, and the transparency column is empty in all 1440 rows, so that gate was never evaluated and is not claimed here. Under embeddability and the 0.05 threshold, 32 of the 72 labels pass. Figure 1a shows the degree-1 undenoised block of this grid in the plane of the two prediction criteria, with the 0.05 gate encoded by marker fill. 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312

Four properties of this grid bound what it can support. (i) The rollout metrics are computed on the same 2018–2019 data the model was fitted to and are therefore in-sample; a separate held-out block (240 fits: 10 seeds $\times$ 3 libraries $\times$ 2 estimators $\times$ 4 fit-year $\rightarrow$ eval-year directions, regen/results/holdout/) splits the two training years to test whether the library ordering survives. (ii) The constrained and sr3 estimators produce bit-identical fits in all 360 seed-cells, so the ladder holds 54 distinct configurations under 72 labels and pooled per-library averages double-weight that estimator. (iii) Every degree-2 configuration is non-embeddable, so the embeddability gate is exactly the degree-1 filter here and carries no independent information. (iv) All rollout horizons are supplied and audited in control steps rather than calendar days; the reported ladder is the step-indexed recomputation used by every analysis here. 313 314 315 316 317 318 319 320 321 322 323

2.6. MPC formulation and stage cost 324

The explicit form of Equation (2) allows it to be embedded analytically in a finite-horizon optimal control problem [4]: 325 326

$$\min_{u_0, \dots, u_{N-1}} \sum_{k=0}^{N-1} [\ell_{\text{res}}(u_k) + \ell_{\text{corr}}(x_k)] + \ell_{\text{corr}}(x_N) \quad \text{s.t.} \quad x_{k+1} = \Xi \Theta(x_k, u_k, d_k), \quad 327$$

with $N = 20$ steps (5 h). The corridor term is $\ell_{\text{corr}} = 30 b_T^2 + 3 \cdot 10^{-4} b_c^2 + 8 b_\varphi^2$, where b_T , b_c and b_φ are excursions from a diurnally varying productive band: temperature in $[15 + 3w, 19 + 5w]^\circ\text{C}$, CO_2 above $400 + 300w$ ppm and relative humidity below 85 %, with $w \in [0,1]$ a smooth day weight. Violations are penalised softly so the problem stays feasible; temperature is hard-bounded to $[12,35]^\circ\text{C}$, ventilation to $[0,0.4]$ and 328 329 330 331 332

the other controls to $[0,1]$, and a move-suppression term penalises control increments. The problem is solved by an interior-point method (IPOPT [38] through do-mpc [39]), whose initial guess is set once from the measured initial state and carried forward by the solver itself; no controller-specific warm start is supplied. The budget is 100 solver failures, shared by every solver-based controller. When a solve fails the step falls back to a fixed action (boiler 0.3, thermal screen closed, all else zero) and the failure counts against that budget; a run that exhausts it is excluded (Section 2.8).

Two resource-weight variants were evaluated in ℓ_{res} . The default implementation uses $20u_{\text{boil}} + 10u_{\text{lamp}} + 2u_{\text{CO}_2}$, a ratio of 1:0.5:0.1. Dividing the logged cost columns by the logged actuator sums in regen/results/final/main.csv gives marginal costs per unit of accumulated normalised control of 0.00293, 0.00870 and 0.00135 EUR for boiler, lamps and CO_2 , respectively, or 1:2.97:0.46. The primary priced formulation therefore uses $7.22u_{\text{boil}} + 21.44u_{\text{lamp}} + 3.33u_{\text{CO}_2}$, scaled so the total at $u = (1,1,1)$ is 31.99 against 32.0 under the default formulation. The corridor term is unchanged, isolating the relative actuator weights while preserving the overall resource-corridor balance.

The price-aligned formulation is used for the main closed-loop comparison and the mechanism block. The supporting blocks on unseen seasons, shortened horizon, bootstrap-draw count, design and coefficient sensitivity, adaptation, the safety guard and fault injection were generated with the default formulation and are labelled accordingly. They should be interpreted as robustness analyses of that formulation; coefficient and threshold sensitivity have not been re-evaluated under the price-aligned weights.

2.7. Rule-based reference and its tuning

To avoid an unsearched baseline, the heuristic was given a 16-trial setpoint-search budget (Table 3). Sixteen configurations were drawn i.i.d. uniformly from a fixed generator (default_rng(20260810)) and scored alongside the hard-coded configuration, which enters the search as trial 0, giving 17 evaluated configurations. Selection used the mean of J over the training seasons 2018–2019 only; the test seasons play no part in it. The heuristic is deterministic, so one run per configuration and season suffices and there is no run-to-run dispersion.

Table 3. Rule-based setpoint search. Ranges, hard-coded values and the selected configuration. Source: regen/results/n2_tune/tune_rb_n2.csv.

Setpoint	Range searched	Hard-coded	Selected
temp_setpoint_day (°C)	17.0–23.0	19.5	22.92
temp_setpoint_night (°C)	14.0–20.0	16.5	14.38
co2_day (ppm)	400–1200	800	828.26
lamp_rad_sum_limit	5.0–25.0	10	7.62
Training J (mean 2018–2019, EUR/m ²)	—	−1.937	+0.920
Test J (mean 2020–2023, EUR/m ²)	—	−1.226	+2.261

Tuning is worth +3.49 EUR/m² over the four test seasons, concentrated in the two adverse ones, and moves the heuristic towards a wider diurnal swing (warmer by day, cooler by night) with a lower lamp threshold; it is bought with constraint pressure, the seasonal violation count rising from 2813 to 4339 steps. Only 2 of the 16 draws beat the hard-coded setpoints, placing the stock configuration near the 88th percentile of the searched space: it was not arbitrary, merely unsearched.

2.8. Statistical treatment

Dispersion is generated by re-collecting the identification data in every repeat, so each seed identifies a different surrogate; for the reinforcement-learning controllers the seed also re-runs training. In the main closed-loop comparison twenty seeds were run for every surrogate-based and learned controller; the supporting blocks use fewer, and their

n is stated with each. Two exceptions are quoted wherever they appear: the neural MPC has 20 seeds under the default objective but only 10 under the priced one, and the idealised controller loses one whole season to the exclusion rule below. The heuristic is deterministic: the stock version was nevertheless repeated on all 20 seeds in the main wave, returning 20 identical values per season, while the tuning wave ran the stock and tuned versions once per season, so the tuned reference has $n = 4$ and no seed axis at all.

Three accounting rules precede any average. First, several waves were resumed after interruptions and their files overlap, so rows are deduplicated on (method, seed, test_year) — or on the block-specific factor key — before aggregation; the deduplicated priced pool contains 600 rows. Second, a run is excluded when it was truncated *and* exhausted the solver-failure budget, which removes 20 of the 800 rows of the canonical main table, all oracle_mpc in 2022. Third, seasons terminated by the simulator rather than the solver are retained as outcomes (42 such seasons in the canonical table), and every headline figure is recomputed three ways — as-is, over complete seasons only, and normalised by the realised season fraction — with any rank change reported.

Paired comparisons use the Wilcoxon signed-rank test [33] with Holm correction [34]. Because Holm's adjustment depends on the family, the family size is stated with every adjusted p -value, and adjusted values from different families are not comparable. One departure is fundamental rather than presentational: because both rule-based references are deterministic, comparisons against either are **one-sample signed-rank tests of the differences against a constant**, not genuinely paired tests, and the reference carries no sampling uncertainty. The same limit binds any test that treats a deterministic controller as the sample: at $n = 4$ the minimum attainable two-sided p is 0.125, so such a comparison can never reach significance whatever the effect size, and its point estimate is reported without a significance claim. An across-season spread quoted for a deterministic controller is labelled as such and is not an error bar.

Effect sizes accompany every p -value, with medians preferred to means wherever the distribution is skewed, and bootstrap confidence intervals [35,40] for aggregate quantities. This is deliberate: the data contain a contrast significant at $p = 0.012$ for a difference of 0.0066 EUR/m², and another whose mean is nine times its median.

2.9. Reproducibility, provenance and its limits

Random number generation is pinned per run coordinate: one helper derives the key from (experiment, controller, seed) and seeds Python, NumPy and torch through a single implementation, BLAS thread counts are fixed, and the 20 seeds are declared in the frozen configuration rather than passed on the command line. Every result row carries a config_hash (637c6b535a9e for all waves reported here) and a git_sha. The hash is identical across the default- and priced-objective waves and so cannot by itself discriminate between them, and the per-row variant fields only partly can: horizon and max_solver_failures are written by the main runner but by none of the supporting blocks, and objective only by waves generated after that flag was added — it is absent from the canonical main table, which is default-objective by construction. The identifying key is therefore the results tree and block together with these fields where present; git_sha is unknown in a few early resumed files.

A self-test runs the pipeline twice under identical seeds and compares SHA-256 digests of the training states and actions, the ensemble coefficients, the STLSQ coefficients, the neural-network weights and the closed-loop trajectory, together with the closed-loop J to twelve decimal places: all seven digests were identical on the pinned stack, and all nine with the reinforcement-learning policies added. Its coverage is narrower than the study: the closed-loop leg is a single surrogate-MPC rollout of one recipe on one season

under the default objective, and no supporting block, neither the identification ladder nor the replay block, is exercised.

One caveat qualifies every reproducibility statement here, and it is a limit on what was measured rather than a measurement. The self-test establishes bit-identity *within* one environment. Nothing in the results tree establishes it *across* environments, and no such claim is made: the study was executed in a single environment, so the cross-environment comparison was never run and its magnitude is unknown. The provenance record makes the gap explicit rather than hiding it: although repro.py can emit an environment fingerprint (--fingerprint), the field is absent from all twenty run manifests under regen/results and the generated provenance map prints env_hash: n/a. Waves are separated only by git_sha — eleven distinct values across those twenty manifests at one common config_hash — which identifies the code but not the machine it ran on.

What the tree does show is that closed-loop margins move at a magnitude the identifying fields do not capture, even for a controller with no stochastic element. The hard-coded heuristic returns a four-season mean of -1.2061 EUR/m² in the main harness and -1.2264 in the tuning harness at identical config_hash, the per-season discrepancy growing monotonically over the four seasons (0.003, 0.013, 0.027, 0.038 EUR/m² for 2020–2023). A deterministic controller re-run on the same season should return the same number; that it does not is chaotic amplification of last-bit floating-point differences through a 5760-step nonlinear loop, which is the same mechanism that makes agreement across operating environments implausible *a priori*. The rule adopted here is therefore that a whole wave is regenerated in one environment and is never mixed cell-by-cell with another for closed-loop numbers. Because the environment itself is not recorded in the manifests, this provenance scheme cannot support a cross-platform guarantee; the reproducibility claim is within-environment only.

3. Results

Throughout, the economic performance indicator (EPI) is the seasonal margin in EUR m⁻², and constraint pressure is the seasonal count of simulation steps outside the productive envelope. Result rows were deduplicated on (method, seed, test_year) before averaging, because several waves were resumed after interruptions and the files overlap; runs in which the solver exhausted its failure budget were excluded. All waves reported here carry the same config_hash (637c6b535a9e), which therefore identifies the code base but *not* the experimental variant; variants are identified by the per-row objective and horizon fields and by the results tree. Two objectives appear: the original stage cost, and the *priced* stage cost, in which the actuator weights were made consistent with the prices the economic criterion itself uses. Section 3.3 uses the priced objective for every controller that optimises a surrogate stage cost, which is every model-predictive controller except the full-model planner — that one maximises the simulator's own economic return and has no weights to reprice. The blocks generated by the supporting harness run under the original objective, and this is stated wherever it matters. Finally, closed-loop trajectories amplify platform-level floating-point differences over 5760 steps: the deterministic rule-based controller evaluates to -1.2061 EUR m⁻² in the main harness and -1.2264 in the tuning harness under identical code and config_hash, so waves are compared within one environment and never cell-by-cell across environments.

3.1. One-step and multi-step criteria rank the feature libraries oppositely

The identification ladder evaluated 72 labelled configurations (three feature libraries × two polynomial degrees × four optimizers × three denoisers) at 20 seeds each, 1440 fits with no duplicates and a complete grid. Table 4 gives the degree-1, undenoised block and Figure 1 plots the two prediction criteria and the closed-loop outcome for all three

libraries. Panels 1a,b use the two sparse estimators in that block ($n = 40$ fits per library): open symbols mark fits that fail the 0.05 divergence gate and crosses show the per-library median and IQR. Panel 1c holds degree, denoiser, ensemble estimator and threshold fixed, changes only the feature library, and adds the 17-feature probe that deletes the bi-linear boiler term ($n = 80$ runs per controller).

Adding physically motivated features monotonically inflates the regression condition number, from $\kappa = 8.21$ (raw) through 24.52 (physics without cross terms) to 53.43 (full physics), and monotonically degrades free-running prediction: the median 24-h rollout RMSE of the inside-air temperature rises from 3.31 to 7.11 to 30.84 °C, and the fraction of diverging rollouts from 0.000 to 0.011 to 0.130. The mean rollout error of the physics library, 1356.7 °C, is an artefact of divergent realisations (upper rollout quartile 2360.1); the median is the only defensible summary. κ depends on the library and the denoiser, not on degree or optimizer. The relation is equally clean at fixed optimizer, which matters because the closed-loop comparison below holds the optimizer fixed: under the ensemble estimator alone the median rollout errors are 2.67, 10.99 and 24.27 °C and the diverged fractions 0.0000, 0.0208 and 0.0767.

This is the whole of what conditioning is claimed to predict in this work. It orders the three libraries on open-loop multi-step stability, monotonically, and the ordering reproduces on a held-out year (below). It does *not* transfer to the closed loop. The same three libraries, compared one factor at a time — ensemble estimator, threshold 0.05, only the library changed — return +4.32, +0.28 and +2.75 EUR m⁻² over four test seasons (Table 6), so the worst-conditioned library outscores the middle one by a factor of ten and the series in κ is non-monotone in closed loop. Nothing in this subsection is therefore an explanation of the economic result: the mechanism that does account for it is actuator-pathway survival, developed in Section 3.5.

The reversal against one-step accuracy is clean only for the two sparse estimators (STLSQ and ensemble, lower panel of Table 4): there the one-step ordering is strictly physics < physics_no_cross < raw (1.675 < 1.728 < 1.862 °C), the exact inverse of every multi-step ordering. Pooled over all four optimizer labels the one-step ordering is not strict — the raw library also beats physics_no_cross on one-step error (2.137 vs. 2.174). The supportable statement is therefore: *the library with the best one-step accuracy is the worst on every multi-step metric*, and the strict inversion holds under sparse estimation. Two properties of the grid qualify the counts: the constrained and sr3 labels are bit-identical in all 360 seed-cells, so 72 labels cover 54 distinct fits and the pooled upper panel of Table 4 double-weights that estimator; and embeddability coincides exactly with the degree-1 filter here (36 of 36 against 0 of 36), so it is not an independent screen. Of the 72 labels, 32 pass both pre-registered gates (embeddable and diverged fraction ≤ 0.05).

The pre-specified frozen recipe (physics_no_cross/degree 1/ensemble/no denoising) passes both gates (diverged fraction 0.0208), but it ranks 8th of 72 on one-step error and 59th–60th on every multi-step metric. The comparator raw/degree 1/STLSQ/no denoising dominates it simultaneously on all three pre-registered axes: rollout RMSE 2.684 vs. 10.496 °C, diverged fraction 0.0000 vs. 0.0208, active terms 19.9 vs. 28.25. The frozen recipe is better on exactly one metric, one-step accuracy, which was never a gate. The full physics library at the same degree and estimator is worse still on the pre-registered axes — 63rd of 72 on median rollout error, 64th on divergence — and it is one of the 40 labels that *fail* the divergence gate (0.0767 > 0.05), so the open-loop procedure would have rejected it outright.

That last fact bounds what the selection rule may be claimed to do, and the bound belongs here rather than in the discussion. The rule identifies the winner: raw is first on multi-step stability and first in closed loop. It does not order the two losers. In closed loop the library the gate rejects scores +2.75 EUR m⁻² and the library the gate passes scores

+0.28 (Table 6) — the reverse of their open-loop ranking. The supportable claim is that multi-step stability is a better selection criterion than one-step accuracy and picks the best of the three libraries, not that it ranks libraries.

A held-out check (240 fits, 10 seeds $\times$ 3 libraries $\times$ 2 sparse optimizers $\times$ 4 fit→evaluate directions over 2018–2019) reproduces the library ordering in both regimes, with held-out/in-sample median ratios of $\times 1.04$ (raw), $\times 1.27$ (physics_no_cross) and $\times 0.99$ (physics). It does not, however, isolate held-out status as the operative factor: pooled over fit year, median rollout error on 2018 versus 2019 data is 2.80/2.61 for raw, 16.94/2.70 for physics_no_cross and 27.19/15.85 for physics, and physics_no_cross narrowly beats raw in the 2018→2019 direction. The honest reading is that the raw library is stable irrespective of evaluation year whereas the physics libraries are strongly year-dependent.

Table 4. Identification ladder, degree-1 configurations without denoising. Upper panel pools the four optimizer labels ($n = 80$ fits per library); lower panel restricts to the two sparse estimators, STLSQ and ensemble ($n = 40$). Rollout RMSE is for the inside-air temperature at a 96-step (24-h) horizon. Source: regen/results/ladder_rerun/.

Library	κ	Active terms	One-step RMSE (°C)	Rollout median (°C)	Rollout IQR	Diverged fraction
<i>All four optimizer labels</i>						
raw	8.21 ± 0.22	12.95	2.137	3.31	[2.67, 4.43]	0.000
physics_no_cross	24.52 ± 0.94	18.66	2.174	7.11	[4.24, 10.99]	0.011
physics	53.43 ± 2.40	26.74	2.022	30.84	[24.08, 2360.14]	0.130
raw	—	20.05	1.862	2.67	—	0.0000
physics_no_cross	—	28.18	1.728	10.58	—	0.0204
physics	—	36.98	1.675	24.27	—	0.0758

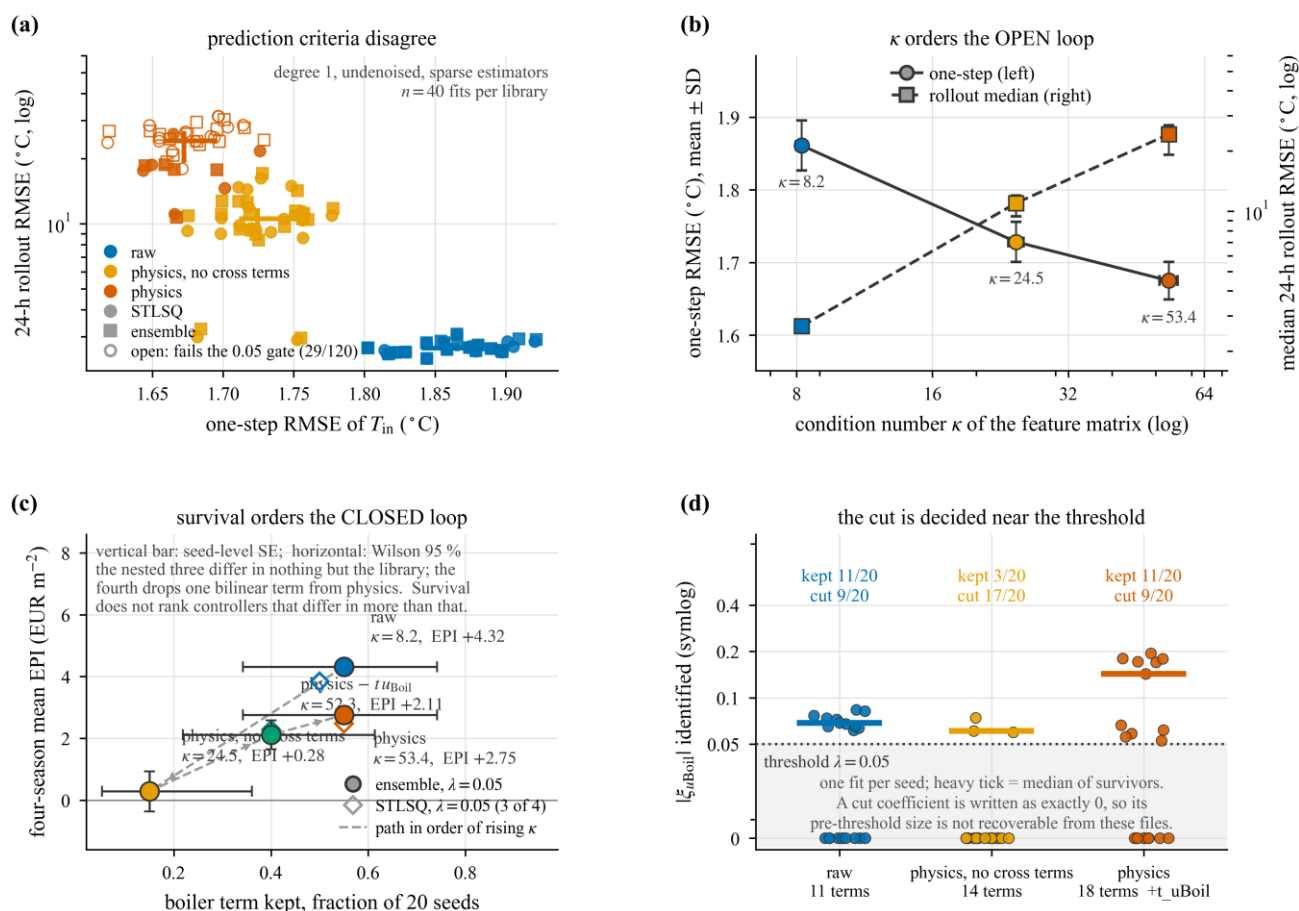


Figure 1. Feature-library choice under open- and closed-loop criteria. (a) One-step versus 24-h rollout RMSE. (b) The same prediction metrics versus the regression condition number. (c) Four-season simulated margin and boiler-term survival for the nested libraries and the 17-feature deletion probe. Error bars show SD.

3.2. Tuning the rule-based reference

The rule-based controller was tuned under the same 16-trial budget the reinforcement-learning agents are recorded as having received — 16 configurations drawn i.i.d. from a fixed generator, evaluated alongside the hard-coded configuration entered as trial 0, so 17 configurations were scored in all — and selected on the training seasons 2018–2019 only, the test seasons taking no part in selection. Table 5 reports the outcome. The predictive controllers received no hyper-parameter search of any kind (Section 2.3), so the budget is matched to the reinforcement-learning family alone.

Tuning is worth +3.49 EUR m^{-2} over four test seasons ($-1.23 \rightarrow +2.26$), and the gain is concentrated in the two adverse seasons, 2021 (+4.99) and 2023 (+5.79). The selected solution is agronomically interpretable — a wider diurnal swing (day setpoint 22.9 instead of 19.5 $^{\circ}\text{C}$, night 14.4 instead of 16.5 $^{\circ}\text{C}$) with a lower supplementary-lighting threshold — and it is bought with constraint pressure (2813 $\rightarrow$ 4339 violation steps per season). Two facts qualify the exercise. Only 2 of the 16 random draws beat the hard-coded setpoints on the training score, placing the stock configuration at roughly the 88th percentile of the search space: it was not arbitrary, merely unsearched. And the heuristic is deterministic, giving $n = 4$ with no run-to-run dispersion; the spread across seasons is not an error bar.

Table 5. Tuning the rule-based reference under the reinforcement-learning agents’ budget (16 random trials plus the hard-coded configuration as trial 0; selection on 2018–2019 only). EPI in EUR m^{−2}; one deterministic run per season. Source: regen/results/n2_tune/tune_rb_n2.csv.

Setpoint	Range searched	Hard-coded	Selected
Day temperature (°C)	17.0–23.0	19.5	22.92
Night temperature (°C)	14.0–20.0	16.5	14.38
Daytime CO ₂ (ppm)	400–1200	800	828.3
Lamp radiation-sum limit	5.0–25.0	10	7.62
Test season	2020	2022	Mean Violations
Hard-coded	+4.63 −6.98	+1.30 −3.85	−1.23
Tuned	+6.29 −1.99	+2.81 +1.94	+2.26
Gain	+1.66 +4.99	+1.50 +5.79	+3.49 +1526

3.3. Simulation-based four-season economic comparison

Fifteen controllers were compared on the test seasons 2020–2023 (Table 6; Figure 2). Every *surrogate-based* model-predictive controller ran under the priced objective at horizon 20. The reinforcement-learning controllers and the heuristic have no surrogate stage cost, and the full-model planner optimises the simulator’s own economic return directly rather than a weighted surrogate cost, so none of the three was repriced and all are quoted from the canonical wave. Two of the fifteen — *sindy_mpc_phys_ens* and *sindy_mpc_phys* — carry the full physics library at the frozen recipe’s threshold of 0.05 and complete the one-factor library series; they were run under the priced objective only, and they are the only two rows of the table with no counterpart under the original stage cost.

The raw-library SINDy-MPC leads with an ensemble optimizer (+4.32 EUR m^{−2}) and with STLSQ (+3.83), ahead of the two low-threshold physics_no_cross variants (+3.09, +3.08), the two full-physics variants (+2.75, +2.48), the tuned heuristic (+2.26), the on-policy re-identified variants (+1.66, +1.37), PPO [30] (+0.45), the frozen recipe (+0.28), the stock heuristic (−1.21), the full-model planner (−2.65, three seasons only), the neural-surrogate MPC (−3.82, 10 seeds) and SAC [31] (−4.15). The ranking of the nine SINDy controllers is therefore not ordered by feature library: the two full-physics controllers sit between the low-threshold physics_no_cross variants and the frozen physics_no_cross recipe, and *sindy_mpc_phys_ens* outranks *sindy_mpc_conf* by four places on a one-factor change of library alone — same degree, same denoiser, same optimizer, same threshold.

Figure 2 places one marker per controller on the plane (EPI ↑, violation steps ↓), with the non-dominated controllers joined by a step line. The advantage is not Pareto-dominant. On that plane the non-dominated set, recomputed over all fifteen controllers, still has exactly five members — *sindy_mpc_raw_ens* (+4.32, 4130), *sindy_mpc_lowthr* (+3.09, 3692), *sindy_mpc_dense* (+3.08, 3686), *sindy_mpc_dense_dagger* (+1.37, 3615) and *ppo* (+0.45, 1330). Neither new controller enters it: *sindy_mpc_raw_ens* dominates *sindy_mpc_phys_ens* (+2.75, 4193) and *sindy_mpc_phys* (+2.48, 4174) on *both* axes, higher margin at fewer violation steps, so the full physics library buys nothing that the raw library does not already provide. PPO attains roughly one third of the constraint pressure of the leader and is the only controller on the front that trades margin for safety at that rate. The tuned heuristic is *off* the front under the priced objective, dominated on both axes by *sindy_mpc_dense*. Front membership of *sindy_mpc_dense_dagger* is marginal: 7 of its 80 runs terminated early, and over completed runs only its violation mean rises to 3657 against 3683 for *sindy_mpc_dense*, narrowing the gap from 71 to 26 steps.

Paired tests against the tuned reference (Table 7, Holm correction over a pre-declared family of 15 controllers) place four controllers significantly above it, all of them SINDy-MPC: both raw-library variants (Δ median +2.85 and +2.83, 75/80 and 68/80 wins) and

both low-threshold physics_no_cross variants (+0.74, +0.69). Three controllers are not separable from the reference and must be described that way rather than as below it: *sindy_mpc_phys_ens* (Δ mean +0.49, median +0.25, 41/80, $p_{\text{Holm}} = 0.176$), *sindy_mpc_phys* (+0.21, +0.29, 44/80, 0.334) and *sindy_mpc_conf_dagger* (−0.60, −0.84, 31/80, 0.334). Every other controller in Table 7 is significantly below. The fifteenth member of the correction family, omitted from the table, is the stock heuristic re-evaluated inside the tuning harness: at $n = 4$ the signed-rank test cannot fall below $p = 0.125$, so it is non-significant (Δ mean −3.49, 0/4, $p_{\text{Holm}} = 0.334$) even though its point estimate matches the significant rule_based row. Because the tuned heuristic is deterministic with one run per season, its value is a constant reused across seeds: this is a one-sample signed-rank test against a constant, and the reference carries no sampling uncertainty. In a separate paired comparison between the two raw variants the test is null (Δ median −0.011, $p = 0.60$) — the library, not the optimizer, carries the effect. One caution on reading these levels: *sindy_mpc_dense* and *sindy_mpc_lowthr* are effectively one controller under the priced objective (per-run mean $|\Delta| = 0.026$, four-season means 3.0822 vs. 3.0888), yet a paired test between them returns $p = 0.012$. Significance here is not magnitude, and the Holm family size is a reporting choice that shifts every corrected level.

The comparison of default and price-aligned stage costs is itself a result. Matched on seed and season, the priced objective improves *both* axes for all eight MPC controllers for which both objectives were run, significantly in every case. The two full-physics controllers are not among those eight and exist only under the priced objective, so nothing here is claimed about their objective sensitivity. The magnitude differs by an order of magnitude between the two libraries that were run twice: +2.74 and +2.68 EUR m^{−2} for *sindy_mpc_dense* and *sindy_mpc_lowthr*, with violations falling from ~6005 to ~3690, against +0.25 and +0.21 for the raw-library variants — a ratio of 10.9 to 12.8 depending on which pair is taken. Consequently the gap between the raw library and the best physics_no_cross controller narrows from +3.66 to +1.23 EUR m^{−2}. The direction is retained, but the magnitude is highly sensitive to whether resource weights match the evaluation prices. Against the full physics library the priced gap is +1.56 (4.3173 against 2.7540), measured under one objective only and therefore not decomposable in the same way.

Table 6. Four-season economic comparison, EPI in EUR m^{−2} (mean $\pm$ SD over runs; n runs = seeds $\times$ seasons). MPC rows use the priced objective at horizon 20. The feature library of each SINDy-MPC row is given in the second column: raw is the 11 measured signals, p_nc adds three nonlinear transforms of the states (physics_no_cross), phys adds four bilinear state

Controller	Lib.	2020	2021	2022	2023	Four-season mean $\pm$ SD	n	Viol. steps
<i>sindy_mpc_raw_ens</i>	raw	9.10 $\pm$ 0.36	−1.56 $\pm$ 0.58	4.62 $\pm$ 2.70	5.10 $\pm$ 0.39	+4.32 $\pm$ 4.08	80	4130
<i>sindy_mpc_raw</i>	raw	8.40 $\pm$ 2.24	−1.71 $\pm$ 0.70	4.04 $\pm$ 3.40	4.61 $\pm$ 1.59	+3.83 $\pm$ 4.23	80	4466
<i>sindy_mpc_lowthr</i>	p_nc	6.58 $\pm$ 1.25	0.27 $\pm$ 1.73	3.22 $\pm$ 1.35	2.29 $\pm$ 1.48	+3.09 $\pm$ 2.71	80	3692
<i>sindy_mpc_dense</i>	p_nc	6.58 $\pm$ 1.23	0.25 $\pm$ 1.71	3.21 $\pm$ 1.34	2.29 $\pm$ 1.45	+3.08 $\pm$ 2.70	80	3686
<i>sindy_mpc_phys_ens</i>	phys	6.35 $\pm$ 2.12	−1.91 $\pm$ 0.87	3.07 $\pm$ 2.34	3.51 $\pm$ 1.44	+2.75 $\pm$ 3.47	80	4193

sindy_mpc_phys	phys	5.82 ± 2.87	−2.02 ± 1.01	2.89 ± 2.76	3.22 ± 1.98	+2.48 ± 3.62	80	4174
rule_based_tuned	—	6.29	−1.99	2.81	1.94	+2.26	4	4339
sindy_mpc_conf_dagger	p_nc	4.56 ± 3.09	−1.73 ± 2.43	2.23 ± 3.74	1.57 ± 2.52	+1.66 ± 3.71	80	4138
sindy_mpc_dense_dagger	p_nc	4.74 ± 2.59	−1.59 ± 1.09	1.70 ± 1.78	0.63 ± 1.37	+1.37 ± 2.89	80	3615
ppo	—	3.36 ± 1.59	−2.10 ± 1.03	1.72 ± 1.19	−1.17 ± 3.40	+0.45 ± 2.97	80	1330
sindy_mpc_conf	p_nc	3.88 ± 4.23	−3.16 ± 1.34	−0.31 ± 3.42	0.72 ± 3.46	+0.28 ± 4.11	80	4755
rule_based (stock)	—	4.63 ± 0.00	−6.97 ± 0.00	1.33 ± 0.00	−3.81 ± 0.00	−1.21 ± 4.52	80	2810
oracle_mpc	—	2.77 ± 0.08	−6.76 ± 0.09	—	−3.95 ± 0.06	−2.65 ± 4.03	60	3321
nn_mpc	phys*	−1.78 ± 1.95	−5.94 ± 1.99	−4.63 ± 1.82	−2.93 ± 1.73	−3.82 ± 2.42	40	2818
sac	—	−2.43 ± 1.33	−6.13 ± 1.47	−2.99 ± 1.61	−5.05 ± 1.20	−4.15 ± 2.05	80	1689

Notes. *nn_mpc is a dense neural surrogate fitted on the physics feature variant. It has no sparsity threshold and no identified coefficients, so the boiler-term survival used throughout Section 3.5 is undefined for it (xi_uboil is empty in all 40 of its rows, which must not be read as zero survival) and it is excluded from every survival comparison. oracle_mpc has no 2022 season: all 20 runs exhausted the common solver failure budget and were excluded, so its mean spans three seasons and is not directly comparable (see Section 3.6). nn_mpc has 10 seeds against 20 for all others. The two phys rows have 20 seeds, zero truncated seasons and zero solver failures. Both heuristics are deterministic: rule_based_tuned was run once per season ($n = 4$), and the stock heuristic was repeated over 20 seeds whose within-season SD is exactly zero. For both, the four-season dispersion entry is across-season spread and not run-to-run error, and the paired tests against them are one-sample tests against a constant. The stock heuristic is quoted from final/main.csv; the same controller evaluates 0.02 EUR m^{−2} lower (−1.23) and three violation steps higher (2813) in the tuning harness, and the tuning gain of Table 5 is computed within that harness, so the two values of this row are never mixed.

Table 7. Paired comparison against the tuned rule-based reference (priced objective). $\Delta =$ controller − tuned in EUR m^{−2}; Wilcoxon signed-rank [33] with Holm correction [34] over a family of 15. The reference is deterministic, so this is a one-sample test against a constant. The family size is a reporting choice and shifts every corrected level: adding the two full-physics controllers raised all thirteen pre-existing levels without moving any of them across 0.05.

Controller	Δ mean	Δ median	Wins/ n	p_{Holm}
sindy_mpc_raw_ens	+2.06	+2.85	75/80	8.3×10^{-11}
sindy_mpc_raw	+1.57	+2.83	68/80	6.5×10^{-7}
sindy_mpc_lowthr	+0.83	+0.74	51/80	7.0×10^{-4}
sindy_mpc_dense	+0.82	+0.69	51/80	7.0×10^{-4}
sindy_mpc_phys_ens	+0.49	+0.25	41/80	0.176 (n.s.)
sindy_mpc_phys	+0.21	+0.29	44/80	0.334 (n.s.)
sindy_mpc_conf_dagger	−0.60	−0.84	31/80	0.334 (n.s.)
sindy_mpc_dense_dagger	−0.89	−0.79	23/80	1.2×10^{-4}
ppo	−1.81	−1.78	11/80	2.3×10^{-10}
sindy_mpc_conf	−1.98	−1.91	29/80	6.8×10^{-5}
rule_based (stock)	−3.47	−3.32	0/80	7.3×10^{-14}

Controller	Δ mean	Δ median	Wins/n	p_{Holm}
oracle_mpc	−4.73	−4.79	0/60	1.8×10^{-10}
nn_mpc	−6.08	−5.52	0/40	2.4×10^{-11}
sac	−6.41	−6.68	0/80	1.1×10^{-13}

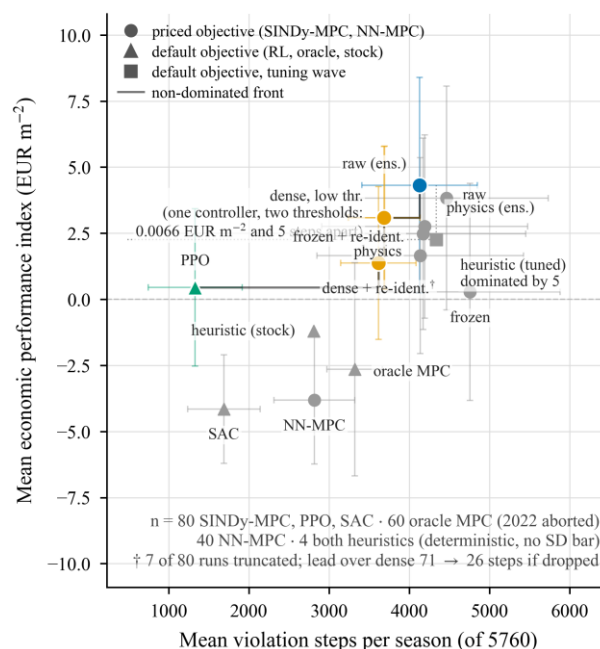


Figure 2. Simulated seasonal margin versus constraint-violation steps under the priced objective, four test seasons and 15 controllers. Whiskers show SD; the step line joins the five non-dominated controllers. Replication counts and exclusions are reported in Table 6.

3.4. Evaluation on four unseen weather seasons

Four weather seasons (2014–2017) that took no part in identification, gate evaluation, selection or tuning were used as an external check (Table 8). These runs use the *default* stage cost; they are therefore not comparable in level with Table 6, only in ordering. The lead holds: the raw-library ensemble controller averages +6.42 EUR m⁻² (SD = 0.95, $n = 32$) against +2.38 for the frozen recipe, +1.07 for the dense physics_no_cross variant and −0.33 for the hard-coded heuristic, and it wins in 32 of 32 seed×season pairs against each. With all differences of one sign at $n = 32$ the signed-rank test is at its exact floor, so these should be read as $p < 10^{-9}$ rather than as measured levels.

The ordering *below* the lead does not carry over. Compared like for like — the same four controllers on 2020–2023 under the same original objective — the dense variant scores +0.34, the frozen recipe −1.26 and the heuristic −1.21; on 2014–2017 the two physics_no_cross variants exchange places and the frozen recipe rises from last to second. Neither full-physics controller was run on these seasons, so this block speaks only to the two libraries it contains. Only the raw library's leadership and the heuristic's last place are invariant across the two sets of seasons, which is consistent with the year-dependence of the physics-informed libraries found in the held-out check (Section 3.1) and a further reason not to read any single ranking of the physics-informed variants as a property of the recipe. Three further caveats: only 8 seeds were run here against 20 in the main wave; the reference is the *hard-coded* heuristic, since the tuned setpoints were never evaluated on 2014–2017; and the margin is again bought with constraint pressure (4385 against 2624 violation steps).

Table 8. Four seasons never used at any stage, *original* objective, 8 seeds. EPI in EUR m^{−2}. Source: regen/results/n5_years/main_n5.csv.

Controller	2014	2015	2016	2017	Mean	SD (n = 32)	Viol.
sindy_mpc_raw_ens	+7.08	+6.25	+5.98	+6.38	+6.42	0.95	4385
sindy_mpc_conf	+3.06	+1.70	+2.18	+2.57	+2.38	4.03	5458
sindy_mpc_dense	+1.63	+0.55	+0.84	+1.27	+1.07	1.74	6226
rule_based (stock)	+0.99	−1.13	−2.22	+1.05	−0.33	1.42*	2624

* The heuristic is deterministic and its within-season SD is exactly 0.000 in all four seasons; the entry is across-season spread, not run-to-run error, and the paired test against it is one-sample, as in Table 6.

3.5. Mechanism

What determines the closed-loop margin is not the conditioning of the regression but whether the actuator pathway survives thresholding. Conditioning does predict one thing cleanly, and only that: open-loop multi-step stability, monotonically in the library (Section 3.1), as reported for ill-conditioned sparse regression [28] and as expected from the identification-for-control literature [19–20]. It does not predict the economics. The three libraries compared one factor at a time (Table 9) put the worst-conditioned basis in the *middle* of the economic ranking, ten times above the middle-conditioned one, so the closed-loop series in κ is non-monotone and no monotone conditioning argument can be made from it. The quantity that does track the ranking is the fraction of seeds in which the boiler-input coefficient $\xi_{u_{\text{boil}}}$ is retained by the sparsity threshold: the two libraries at 0.55 survival score +4.32 and +2.75 EUR m^{−2}, and the one library at 0.15 scores +0.28.

The comparison is one-factor by construction: degree, denoiser, optimizer and threshold (0.05) are held at the frozen recipe's values and only the feature library changes, over four seasons and 20 seeds with zero truncated runs. The same pattern appears in the STLSQ arm, where the raw library gives +3.83 (survival 0.50) and the full physics library +2.48 (survival 0.55); that arm has no four-season physics_no_cross controller at threshold 0.05, but the $\lambda = 0.05$ level of the sparsity sweep is exactly that configuration on season 2020, and it reproduces the anomaly — survival 0.10 and +4.04 EUR m^{−2}, against +8.40 (raw) and +5.82 (physics) on the same season. Reading across harnesses is what the environment caveat of the section preamble warns about, so this is offered as a consistency check on the shape and not as a fourth measurement.

Why the middle library is worse than either neighbour had one candidate explanation, and it has now been tested directly and rejected. The three libraries are nested: raw is the 11 measured signals; physics_no_cross adds saturation vapour pressure, vapour-pressure deficit and screen-attenuated radiation, which are nonlinear transforms of measured states and therefore near-collinear with them; physics adds four bilinear state $\times$ input terms on top, among them $T_{\text{in}}u_{\text{boil}}$. The candidate explanation held that the middle library carries enough collinearity for the threshold to destabilise the *direct* boiler term, while the full library — worse conditioned still — keeps the bilinear term as a detour able to carry the heating pathway once the direct one is cut. That account makes a sharp, falsifiable prediction: deleting exactly the bilinear feature from the 18-feature full library — sixteen of eighteen features, and essentially all of the collinearity, unchanged — should collapse the truncated library onto the middle one, with boiler survival near 0.15 and margin near +0.28 EUR m^{−2}, not leave it near +2.75 and 0.55.

The prediction fails (Table 9, fourth row pair). The 17-feature library (physics_no_tu-boil), refit at both sparse estimators at threshold 0.05 and simulated over the same four seasons $\times$ 20 seeds with zero truncated runs, keeps the full library's conditioning ($\kappa = 52.3$ against 53.4, mean over 20 refits), its open-loop instability (median rollout 24.2 °C,

divergence 0.084) and — decisively — its closed-loop level: +2.11 EUR m⁻² under the ensemble estimator and +2.28 under STLSQ. That is significantly above the middle library (mean paired difference +1.83, 62 of 80 pairs, $p_{\text{Holm}} = 4.4 \times 10^{-6}$ over the four contrasts of this block) and at most slightly below the full library (median -0.18, $p_{\text{Holm}} = 6.4 \times 10^{-3}$ under the ensemble estimator; the STLSQ arm is not separable, $p = 0.089$). Boiler-term survival moves only part of the way down, to 0.40 (8 of 20 replicates; exact McNemar test against the full library $p = 0.55$), and nowhere near the predicted 0.15. The small significant cost of the deletion under one estimator suggests the bilinear term shares the threshold's pressure with the direct term rather than substituting for it. With the detour ruled out, the margin ordering across the four libraries continues to track survival — 0.15 → +0.28, 0.40 → +2.11, 0.55 → +2.75 and +4.32 EUR m⁻² — while *which* feature or correlation pattern makes the threshold sever the direct term in physics_no_cross becomes an open question, stated as such in Section 4.

The controlled evidence for the mechanism is the single-coefficient ablation, and it is the only randomised element in this section. On the frozen recipe, paired over 20 seeds and one season, the boiler coefficient is re-inserted from the same seed's low-threshold fit with nothing else changed, lifting survival from 0.15 to 1.00. Under the priced objective the effect is significant ($p = 5.9 \times 10^{-4}$, $p_{\text{Holm}} = 1.2 \times 10^{-3}$ over the two knock tests, positive in 16 of 20 replicates) but typically small and heavily right-skewed: the median is +0.21 EUR m⁻² and the mean +1.92, nine times the median and carried by a few seeds. All three statements belong together; the median is the defensible summary of the typical case and the mean is the defensible summary of the aggregate, and quoting either alone misrepresents the distribution. Under the default objective the same test gives a median of +3.05 at essentially the same significance, so the direction survives price alignment while its typical magnitude falls by a factor of about 14. Much of the apparent value of the coefficient is therefore attributable to resource weighting rather than to the term alone. Knock-out is not significant under either objective and its median is exactly zero, for the plain reason that the term is already absent in 17 of the 20 baseline seeds: there is usually nothing to remove. Cross-term interaction tests, run on the full physics library at threshold 10^{-6} , are uniformly non-significant and two of three point estimates flip sign between the objectives, so that block supports no direction. In particular it does not test the detour: at 10^{-6} the boiler term is retained in all 20 seeds, so the block removes coefficients from a model that never lost any, which is the opposite of the situation the detour is meant to explain.

Figure 3 combines the within-library dose-response and the randomised single-coefficient ablation. Panels 3a,b place margin and boiler-term survival on the same sparsity-threshold grid; panel 3c shows the seed-paired knock-in effect under each objective, with the median separated from the right-skewed mean. Within one library the association is a dose-response. The sparsity sweep (Table 10 and Figure 3, physics_no_cross, season 2020 only, 20 seeds per level) shows the boiler term dying between $\lambda = 0.03$ (survival 0.95) and $\lambda = 0.06$ (survival 0) in both waves, whose survival profiles differ in one cell only ($\lambda = 0.04$: 0.35 priced against 0.30 original, one seed, and a consequence of the execution-environment drift noted above rather than of the objective). Levels at which at least one seed retains the term average +5.77 EUR m⁻² against +3.47 for levels at which none does, under the priced objective (+3.46 vs. +2.77 under the original). The separation is not monotone under either objective — the worst retaining level (4.04, priced) is below the best non-retaining one (4.61) — so the statement “EPI is monotone in boiler-term survival” is false as written, here as it is across libraries. The priced weights raise EPI at every one of the 13 levels (+0.34 to +2.95) and cut violations at every level.

Table 11 decomposes EPI over the four test seasons by library and by whether the boiler term survived, now over all three libraries. Inside physics_no_cross the gap is

+2.34 EUR m⁻² and inside raw +1.13; inside physics it is −0.22 and not significant. The raw library with the term *dropped* (+3.48) still beats every cell of both physics libraries, including physics_no_cross with the term *kept* (+2.36), so survival does not exhaust the library effect either. This decomposition is observational, not a randomised ablation — survival is a property of the fit, constant across the four seasons of a seed, so the effective sample is 20 seeds per controller and not 80 runs, and the cells differ in composition as well as in the factor of interest: under the priced objective the “kept” cell of physics_no_cross is filled by the three recipes that never lose the term (80 runs each), plus most of one re-identified variant (68 of 80), plus the 12 runs in which the frozen recipe happens to keep it; its “dropped” cell is 68/80 the frozen recipe. Between seeds the boiler association is significant only inside physics_no_cross ($p = 0.0015$ priced, $p = 0.0006$ original) and not inside raw ($p = 0.136$, $p = 0.316$) or physics ($p = 0.178$). The causal estimate remains the knock-in median of +0.21, not the −0.2 to +2.3 spread the three observational splits suggest.

Four limits qualify the whole of this subsection. The randomised ablation is single-season (2020) and single-recipe, so the causal estimate is not measured under the seasonal variation that the rest of the results report. The knock and cross blocks are anchored on different baselines (survival 0.15 against 1.00) and on different libraries, so their deltas are not one ablation family. The library comparisons — including the term deletion, which is a structural edit and not an assigned factor — are randomised in the seed but not in the library: libraries are not assigned, they are chosen, so these are controlled comparisons rather than experiments on the mechanism. And survival is measured on one coefficient of one actuator; whether the same account holds for the lamp and CO₂ pathways was not tested.

Table 9. One-factor library comparison. Degree, denoiser and sparsity threshold (0.05) are held at the frozen recipe’s values; only the feature library changes. Open-loop columns are degree-1 undenoised ladder fits at the matching optimizer ($n = 20$); closed-loop columns are four test seasons $\times$ 20 seeds under the priced objective ($n = 80$, no truncated runs). ξ survival is the fraction of runs retaining the boiler-input coefficient. The p_ntb rows are the 17-feature term-deletion library that tests the bilinear-detour reading (Section text): deleting the one cross term that touches the boiler leaves κ , rollout and divergence at the full library’s level, and the predicted collapse onto the p_nc row does not occur. Sources: regen/results/ladder_rerun/, priced_main/, priced_dagger/, phys_lib/, notuboil/.

Library	Controller	κ	Rollout median (°C)	Diverged fraction	ξ survival	EPI (four-season)
<i>Ensemble estimator</i>						
raw	sindy_mpc_raw_ens	8.21	2.67	0.0000	0.55	+4.32 ± 4.08
p_nc	sindy_mpc_conf	24.52	10.99	0.0208	0.15	+0.28 ± 4.11
phys	sindy_mpc_phys_ens	53.43	24.27	0.0767	0.55	+2.75 ± 3.47
p_ntb	sindy_mpc_notuboil_ens	52.28	24.17	0.0842	0.40	+2.11 ± 3.51
raw	sindy_mpc_raw	8.21	2.67	0.0000	0.50	+3.83 ± 4.23
p_nc	— (not run)	24.52	10.45	0.0200	—	—
phys	sindy_mpc_phys	53.43	24.49	0.0750	0.55	+2.48 ± 3.62
p_ntb	sindy_mpc_notuboil	52.28	23.58	0.0783	0.40	+2.28 ± 3.55

The first three columns are monotone across the three nested libraries and the last two are not: the worst-conditioned library outscores the middle one by a factor of ten in closed loop, and the economic ranking follows ξ survival rather than κ ; the p_ntb row sits between the two physics rows by construction and is excluded from that statement. No four-season controller exists for physics_no_cross under STLSQ at threshold 0.05; the

$\lambda = 0.05$ level of Table 10 is that configuration on season 2020 and gives survival 0.10 at +4.04 EUR m⁻², against +8.40 and +5.82 for the raw and phys STLSQ rows above on the same season. That is a cross-harness comparison and is quoted as a consistency check only.

Table 10. Sparsity sweep within the physics_no_cross library, season 2020, 20 seeds per level. ξ survival is the fraction of seeds retaining the boiler-input coefficient; it and the active-term count are quoted from the priced wave. Identification precedes the closed loop and does not depend on the objective, but the two waves were executed in different environments, so their fits are not bit-identical: 28 of the 260 (λ , seed) cells differ by one or two active terms, level means by at most 0.15, and one cell flips ξ survival — whence the original-objective survival at $\lambda = 0.04$ is 0.30 rather than 0.35. EPI in EUR m⁻². Sources: regen/results/priced_mech/ and regen/results/final/mechanism*.csv.

λ	Active terms	ξ survival	EPI (priced)	EPI (original)	Δ
10 ⁻⁶	53.95	1.00	6.58 ± 1.25	3.68 ± 1.85	+2.90
10 ⁻³	49.60	1.00	6.58 ± 1.23	3.63 ± 1.93	+2.95
0.01	42.65	1.00	6.11 ± 2.52	3.21 ± 3.25	+2.90
0.02	38.80	1.00	6.38 ± 2.01	4.39 ± 1.47	+1.99
0.03	36.10	0.95	6.41 ± 1.82	4.98 ± 1.76	+1.43
0.04	32.15	0.35	4.30 ± 3.51	2.76 ± 3.84	+1.54
0.05	28.10	0.10	4.04 ± 4.14	1.59 ± 5.59	+2.45
0.06	25.45	0.00	3.59 ± 2.96	2.58 ± 4.63	+1.01
0.07	23.05	0.00	4.01 ± 2.37	3.67 ± 2.62	+0.34
0.08	21.30	0.00	3.04 ± 1.09	2.49 ± 1.23	+0.55
0.10	19.50	0.00	3.11 ± 0.36	2.50 ± 0.34	+0.61
0.15	14.40	0.00	4.61 ± 0.93	4.05 ± 0.99	+0.56
0.20	12.15	0.00	2.43 ± 2.49	1.34 ± 3.05	+1.09

Table 11. Mean EPI (EUR m⁻²) by feature library and boiler-term survival, pooled over the nine SINDy-MPC controllers and four test seasons; run counts in parentheses. The physics library was run under the priced objective only. Sources: final/main.csv, n7/main_n7.csv (original objective), the priced pool and phys_lib/.

Library	Original objective		Priced objective		Boiler effect (priced)
	dropped	kept	dropped	kept	
raw	+3.20 (76)	+4.43 (84)	+3.48 (76)	+4.61 (84)	+1.13 (n.s.)
physics_no_cross	−1.42 (88)	+0.15 (312)	+0.02 (80)	+2.36 (320)	+2.34
physics	—	—	+2.74 (72)	+2.52 (88)	−0.22 (n.s.)
Library effect, raw − physics_no_cross	+4.62	+4.28	+3.46	+2.25	
Library effect, raw − physics	—	—	+0.75	+2.09	

Survival is a property of the fit and is constant across the four seasons of a seed, so the effective sample is 20 seeds per controller and not the printed run count. Between seeds the association is significant only inside physics_no_cross ($p = 0.0015$ priced, $p = 0.0006$ original) and not inside raw ($p = 0.136$, $p = 0.316$) or physics ($p = 0.178$). The last of those three nulls is the observation that motivates the cross-term account in the text; it is not a test of it.

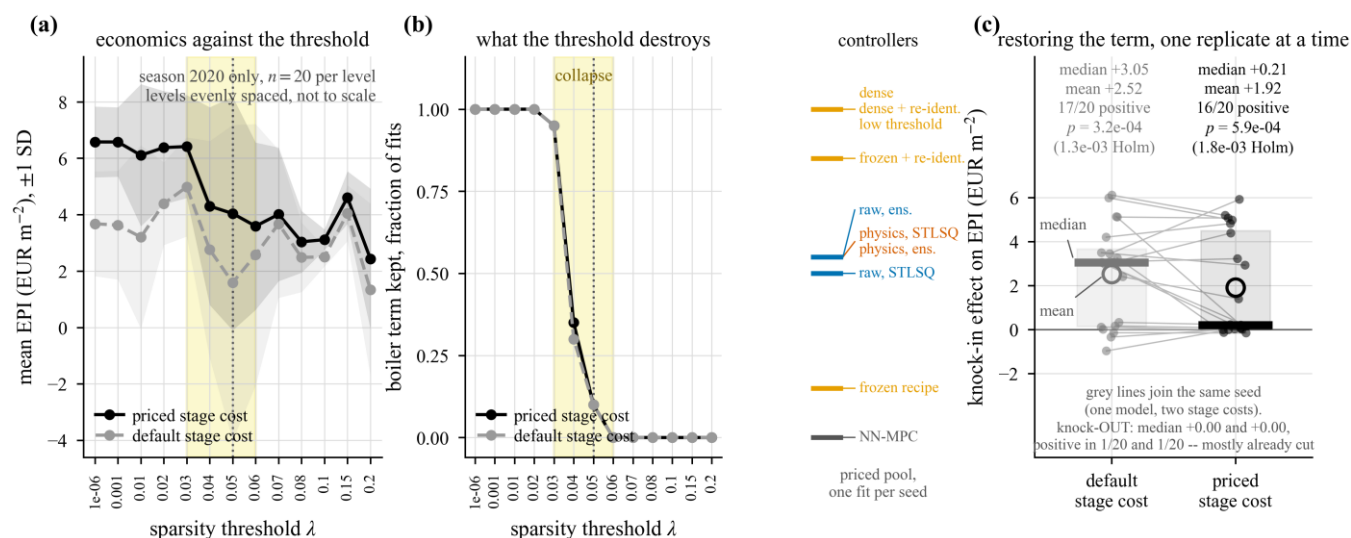


Figure 3. Sparsity threshold and boiler-term mechanism in the physics_no_cross library, season 2020 ($n = 20$ per condition). (a) Mean margin with SD. (b) Boiler-term survival on the same threshold grid. (c) Seed-paired coefficient knock-in; the heavy tick is the median and the open circle the mean.

3.6. Alternative explanations

Model accuracy is not the binding constraint. A planner using the simulator's own dynamics ranks in the lower third (-2.65 EUR m⁻²), and its shortfall cannot be attributed to the price-aligned surrogate cost of Section 3.3; it never optimised the weighted surrogate cost in the first place, but maximised the simulator's own economic return over its horizon. What it does have is a weak optimiser — a warm-started cross-entropy search over a few dozen candidate sequences — so the result is about optimisation quality, not model quality. Its 2022 season is absent from Table 6 because all 20 runs hit the common solver-failure budget of 100, at about 95 % of the season (season fraction 0.936–0.987, mean 0.950). This is a budget artefact, not infeasibility: with the budget raised to 10^5 the season completes on both tested seeds, requiring 140 and 103 failures and yielding -1.12 and -0.56 (mean -0.84). Imputing that mean gives a four-season figure of about -2.20 instead of -2.65 ; both should be quoted, and the correct phrasing is that the full model does not fit the common budget, not that it cannot be solved. Its rank is unaffected either way.

Prediction error along the operating trajectory is small. Replaying both physics_no_cross surrogates along the full-model planner's own closed-loop trajectory (5760 steps, 5 seeds, season 2020) gives one-step temperature RMSE of 1.89 ± 0.09 and 1.82 ± 0.04 °C ($R^2 = 0.82$ and 0.83) and 24-h rollout RMSE of 2.62 and 2.49 °C, with zero diverging rollouts out of 100 at every horizon. The same frozen recipe measures about 10.5 °C on the excitation data of the ladder — roughly four times larger. The surrogate is far more accurate along a smooth closed-loop trajectory than on PRBS-excited identification data, and the ladder number must not be read as its error in operation.

The horizon was never tuned, and correcting for it does not change the ordering. At horizon 8 every MPC improves over horizon 20 on matched seeds and seasons (sindy_mpc_raw_ens $+4.58$ vs. $+4.29$, $\Delta = +0.29$, 33/40; sindy_mpc_lowthr $+2.78$ vs. $+0.67$, $\Delta = +2.11$, 40/40; sindy_mpc_dense $+2.65$ vs. $+0.53$, $\Delta = +2.12$, 40/40; sindy_mpc_conf $+0.90$ vs. -0.40 , $\Delta = +1.30$, 37/40), with no rank change. The size of the gain is inversely related to multi-step prediction quality — $+0.29$ for the raw library against $+1.3$ to $+2.1$ for the physics_no_cross variants — which is exactly the compounding-error mechanism. The two full-physics controllers were not in this block, so this argument is not extended to them. A single-controller horizon sweep confirms the

direction (6.11, 5.13, 3.63, 0.91 EUR m⁻² at $h = 8, 12, 20, 30$, with dispersion growing from SD = 0.88 to 4.03). The asymmetry runs against the MPCs: the learned controllers received 16 tuning trials and the horizon was never tuned for any MPC.

The bootstrap draw of the ensemble optimizer is a real but secondary axis. Resampling the ensemble draw (6 seeds $\times$ 6 draws $\times$ 4 seasons, original objective) leaves the raw-library controller leading in 32 of 36 seed $\times$ draw cells (0.89). Its spread across draws within a seed averages 0.26 EUR m⁻² against 1.11 for the frozen recipe and 2.04 for the re-identified variant, but that mean is not representative: five of the six seeds lie at or below 0.25 and one seed alone contributes 1.18. The controller has one bad basin rather than uniform insensitivity to the draw. This wave disagrees with the canonical 20-seed, 10-draw wave on the frozen recipe's boiler survival (0.00 vs. 0.12) and on the leadership ordering, so the two must be cited separately and never merged.

3.7. Adaptation and safety

Three adaptation modes were compared on the three harder seasons (2021–2023) with 20 seeds $\times$ 10 ensemble draws per condition, 600 runs each, under the original objective. On-policy re-identification — the surrogate is rolled out in closed loop, the visited states are appended to the training set and the model is re-estimated, with no expert queried, so this is not DAgger [41] — gives -0.39 EUR m⁻² against -1.58 for the static model (paired $\Delta = +1.19$, median $+0.41$, 380/600, $p = 5.0 \times 10^{-16}$) and reduces violations from 5424 to 4654. Recursive coefficient adaptation by extended Kalman filter is harmful: -2.95 EUR m⁻² ($\Delta = -1.36$ against static, $p = 7.4 \times 10^{-14}$) with 234 of 600 seasons terminating early. All three are net-negative in absolute terms because 2020, the favourable season, is absent from this block.

The Mahalanobis-distance safety guard [42–43] does not pay. Guarded runs average -2.59 EUR m⁻² against -1.91 unguarded; the paired mean difference is -0.69 but the paired median is $+0.19$ with the guard better in 335 of 600 pairs ($p = 0.0047$), because the guard shortens a heavy left tail while shaving the centre; both statistics must be reported. Violations are *worse* under the guard ($+430$ steps, worse in 529 of 600, $p = 5 \times 10^{-85}$), and 179 of 600 guarded seasons terminate early; restricted to completed seasons the margin difference is -0.33 and not significant ($p = 0.081$). The underlying detector is weak: the area under the ROC curve is 0.663 ± 0.042 for the distance statistic and 0.627 ± 0.058 for ensemble spread [44], barely above chance. The guard's activation count was never recorded, so the natural explanatory variable is unmeasured. This is a negative result and is reported as one.

The residual-based fault supervisor, in contrast, works (Table 12). It improves the margin under all six injected faults (paired over 20 seeds, all $p \leq 1.7 \times 10^{-3}$, and unchanged in that bound under Holm correction over the six) and collapses the dispersion from SD = 1.16–5.48 to 0.29–0.39: it buys predictability more than margin. Violations fall sharply (9131 to 5292 under a dead vent; 5198 to 2660 under a stuck boiler). Two faults remain net-destructive even when supervised. One result deserves plain statement rather than burial: with the lamps dead, the supervised fallback ($+6.33$) outperforms the nominal fault-free controller ($+3.63$), which implies that the nominal MPC over-uses supplementary lighting.

Table 12. Fault injection with and without the residual-based supervisor, season 2020, 20 seeds. EPI in EUR m⁻², mean (SD); paired Wilcoxon over seeds. The fault-free reference is $+3.63$ (SD 1.93). Source: regen/results/final/tables/faults.csv.

Fault	No supervisor	Supervised	Δ	Wins	vs. fault-free
uVent_dead	-18.19 (5.48)	$+0.10$ (0.30)	$+18.29$	20/20	-3.53
uBoil_stuck	-8.41 (5.35)	-3.88 (0.29)	$+4.54$	16/20	-7.50

Fault	No supervisor	Supervised	Δ	Wins	vs. fault-free
rh_offset	+1.27 (3.72)	+4.45 (0.36)	+3.19	20/20	+0.83
uLamp_dead	+3.50 (1.64)	+6.33 (0.29)	+2.82	20/20	+2.70
t_in_stuck	+3.06 (1.18)	+4.42 (0.29)	+1.36	17/20	+0.79
t_in_offset	+3.67 (1.16)	+4.47 (0.39)	+0.80	17/20	+0.84

3.8. Sensitivity to design parameters and to prices

Figure 4 separates two sensitivity questions: panel 4a shows the run-level response to coefficient perturbation, whereas panel 4b re-scores fixed trajectories over a fruit-price $\times$ energy-price grid. Perturbing the identified coefficients degrades the closed loop non-uniformly (Table 13). Mean EPI falls from +4.11 at a 2 % perturbation to −8.13 at 20 %, a span of 12.24 EUR m^{−2}, while the median falls only from +4.47 to −0.31, a span of 5.72: degradation takes the form of a growing minority of catastrophic runs, not a uniform decline. Only the two largest levels are significant against the smallest on per-seed means ($p = 0.0059$ at 15 %, $p = 0.0020$ at 20 %; $p > 0.6$ below that), and the 2 % $\rightarrow$ 5 % step is already inverted. The canonical coarser grid, which also covers 30 %, gives +1.67, −13.41, −4.77 at 10/20/30 % — a span of 15.08 and non-monotone; the finer rerun stops at 20 %, so that non-monotonicity is unreproduced rather than refuted. The coefficient-sensitivity span is several times larger than the mean differences among the leading controllers in Table 6. Varying the sparsity threshold of the same controller spans 5.00 EUR m^{−2} on this 10-seed rerun but only 1.87 on the canonical 20-seed grid; the two disagree by about 1 EUR m^{−2} even on matched seeds at the two lowest thresholds, so the span of this factor is not well determined and only the direction should be relied on. Violations rise by half between the extreme levels (6167 at threshold 0.01 against 9570 at 0.20) but *not* monotonically: the 0.05 level, at which the boiler term is lost in 9 of 10 seeds, has the fewest violations of the four (5155), and the canonical grid reproduces that dip (6263, 5423, 8440, 9219). Varying the horizon spans 4.76–5.21.

Re-scoring the recorded physical quantities over a 3 $\times$ 3 price grid (fruit price $\times$ energy scale, each spanning a factor of four) yields per-controller spans of 31.5 to 50.2 EUR m^{−2}, median 38.3. Prices therefore dominate design parameters numerically, but the axes are not commensurable — a factor of four in prices against ± 20 % on coefficients — and no claim that “prices matter more than design” is supportable from this comparison. What is supportable is that the ranking is not price-invariant either: two distinct controllers win across the nine price points (the heuristic at six, a re-identified SINDy variant at three). This last statement carries an important restriction: the price grid was re-scored only for the ten controllers of the canonical wave, and *neither raw-library controller is in it*. The price sensitivity of the leading controller of Table 6 has therefore not been measured, and the nine-point winner counts must not be read as a re-ranking of the main comparison.

An objective-scope audit established that the supporting harness for the design and sensitivity blocks uses the default objective in every rollout call. The objective flag is consumed only by the main and mechanism blocks; consequently, a directory label containing “priced” does not make these particular reruns price-aligned, and they are reported here only as default-objective results. The mechanism sparsity sweep and the design threshold factor fit the same model at the same levels, and under the original objective the two agree exactly on all 80 overlapping (level, seed) cells (max $|\Delta| = 0$). The rerun labelled priced, by contrast, disagrees with the genuinely priced mechanism sweep on *all* 40 overlapping cells (mean $\Delta = +1.23$ EUR m^{−2}), while remaining within execution-environment drift of the original-objective values (39 of 40 horizon cells and 36 of 40 threshold cells within 0.2 EUR m^{−2}; the remainder are chaotic amplifications). The coefficient and

threshold sensitivities have consequently never been measured under the priced objective.

Table 13. Coefficient-perturbation sensitivity of a single SINDy-MPC controller, season 2020, 10 seeds $\times$ 4 repetitions, original objective. EPI in EUR m^{-2} . Source: regen/results/priced_design/design_pricedDesign*.csv (deduplicated).

Perturbation	Mean	SD	Median	Min	Early terminations
0.02	+4.11	1.29	+4.47	+1.25	0/40
0.05	+4.03	2.06	+4.35	−1.19	0/40
0.10	+2.96	7.57	+5.40	−34.18	0/40
0.15	−3.86	14.27	+1.12	−42.87	4/40
0.20	−8.13	18.39	−0.31	−42.16	12/40
Span	12.24	—	5.72	—	—

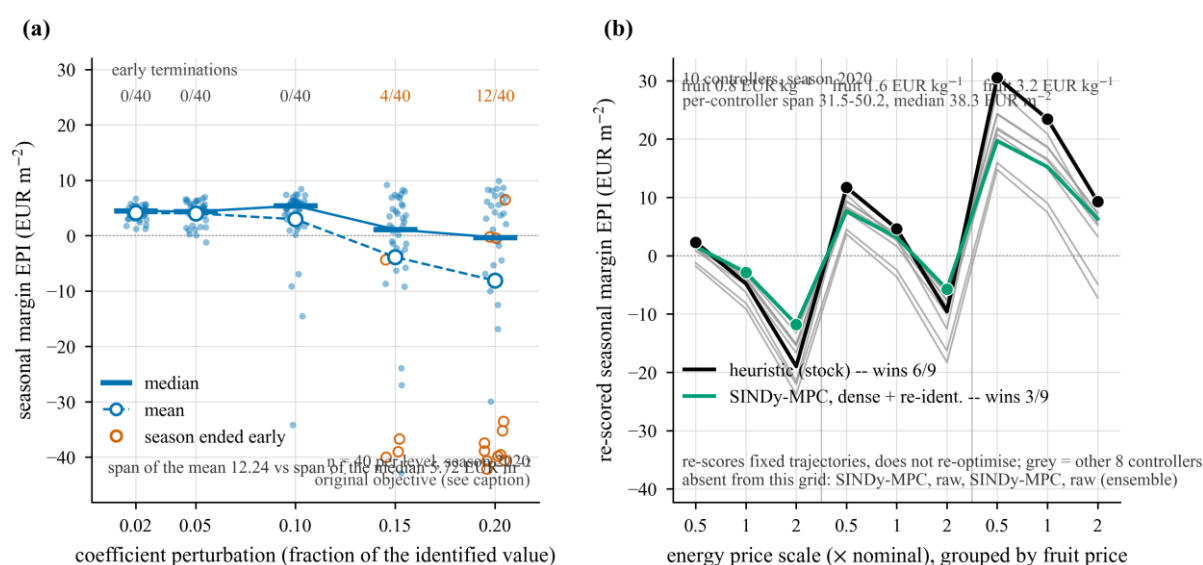


Figure 4. Sensitivity of the simulated seasonal margin. (a) Run-level response to coefficient perturbation under the default objective; the heavy tick is the median and the open circle the mean ($n = 40$ per level). (b) Re-scoring of fixed trajectories over a 3×3 fruit-price $\times$ energy-price grid.

4. Discussion

4.1. Why physically motivated features can hurt, and why the damage is not monotone in how many are added

The physically motivated regressors used here—saturation vapour pressure $e_s(T_{\text{in}})$, vapour-pressure deficit D and screen-attenuated radiation $I(1 - u_{\text{ThScr}})$ —are algebraic functions of variables the raw library already contains: no new excitation, only smooth near-collinear copies of existing columns. Adding them, and then the four bilinear cross terms, raises the condition number of the regression monotonically, and open-loop multi-step stability degrades monotonically along the same direction. Over the ladder's degree-one undenoised block (Section 3.1), $\kappa = 8.2 \rightarrow 24.5 \rightarrow 53.4$, median 24 h rollout error $3.31 \rightarrow 7.11 \rightarrow 30.84$ °C and divergence $0.000 \rightarrow 0.011 \rightarrow 0.130$. One-step error does not order the libraries along that direction at all: it is *best* for the worst-conditioned library (2.02 °C) while the middle library is marginally worse than raw (2.17 against 2.14), so what the ladder establishes is a contrast between the extreme points, not a monotone reversal. Table 14 shows the same contrast on the held-out identification block, where the three libraries score $\kappa = 8.2$, 25.4 and 56.2; that block is a different sample of fits (10

replicates per fit year rather than 20 on both), so the two sets of figures must not be mixed within one comparison.

The open-loop conditioning pattern does not extend monotonically to closed-loop economics. Under the one-factor closed-loop comparison—ensemble optimiser, threshold 0.05, priced objective, four seasons $\times$ 20 replicates, zero truncated runs, the feature library the only thing that changes—the three libraries scored +4.32 (raw), +0.28 (physics without cross terms) and +2.75 EUR m⁻² (full physics). The series is **non-monotone in κ** : the worst-conditioned library beats the middle one by +2.47 EUR m⁻² on average (median +2.39, 63 of 80 paired runs, $p = 1.3 \times 10^{-8}$), and it does so in each of the four seasons taken separately. Conditioning therefore predicts open-loop multi-step stability cleanly but is not a sufficient explanation for the non-monotone closed-loop economics.

What the closed-loop ordering does track is whether the boiler term survives thresholding. The direct coefficient $\xi_{u_{\text{Boil}}}$ is retained in 11 of 20 identification replicates with the raw library, 11 of 20 with the full physics library, and 3 of 20 with the physics library without cross terms; the two libraries at 11 of 20 score +4.32 and +2.75, and the one at 3 of 20 scores +0.28. Restated as a mechanism: what matters is not how well conditioned the regression is, but whether the pathway through which an actuator reaches the controlled state survives sparsification. A surrogate whose boiler column has been thresholded away cannot be steered with the boiler, however accurate it is one step ahead and however well conditioned the fit that produced it.

The controlled evidence for that reading is the single-coefficient ablation. Reinstating the deleted coefficient in the frozen recipe is the only randomised manipulation in the mechanism block—library, scalars, every other term and the MPC construction held fixed, one number changed. Under the priced objective the effect is significant and typically small: positive in 16 of 20 replicates, $p = 5.9 \times 10^{-4}$ (1.2×10^{-3} after Holm correction over the two knock tests; 1.8×10^{-3} if the family is widened to the four mechanism contrasts, as in Figure 3), median +0.21 EUR m⁻² with interquartile range [+0.02, +4.49], mean +1.92 (Figure 3c). The mean is nine times the median because a minority of replicates carry it, so both summaries are required. Under the default, non-price-aligned objective the median is +3.05; the contrast shows that the magnitude, unlike the direction, depends strongly on economic weighting.

Survival gates the outcome; it does not order it, and two measurements say so. Among the two libraries that keep the term, the raw one still wins by +1.56 EUR m⁻² (median +0.97, 67 of 80, $p = 2.5 \times 10^{-9}$)—and it is also the one with the better rollout, so the open-loop criterion of Section 4.2 separates the two survivors where survival does not. And in the decomposition over (library) $\times$ (term retained), the raw library *without* the term (+3.48 EUR m⁻²) still beats the physics library *with* it (+2.36). That split is observational—survival is a property of each fit, not a randomised factor, and it is confounded with recipe—so the conservative reading is that survival marks which fits collapse rather than measuring the whole library effect.

One plausible explanation for why the middle library is worse than *both* of its neighbours is that the library without cross terms carries enough collinearity for the threshold to destabilise the direct boiler term, but—unlike the full library—holds no bilinear $T_{\text{in}}u_{\text{Boil}}$ term able to carry the heating pathway once the direct term is cut: the worse-conditioned library has a detour, the middle one has neither stability nor a detour. Two observational figures were consistent with it, and neither is significant: inside the full physics library, replicates that lose the direct term score no worse than those that keep it (+3.10 against +2.48 EUR m⁻², 9 and 11 replicates, $p = 0.25$), whereas inside the library without cross terms losing it costs +2.26 (+2.20 against -0.06, 3 and 17 replicates, $p = 0.18$).

The term-deletion test rejects this detour explanation. Deleting exactly the bilinear $T_{\text{in}}u_{\text{Boil}}$ feature—the only one of the four cross terms that touches the boiler—from the

full physics candidate set gives a 17-feature library that keeps sixteen of eighteen features and essentially all of the conditioning ($\kappa = 52.3$ against 53.4, mean over the 20 refits). Refit at the same optimiser and threshold and simulated over four seasons $\times$ 20 replicates in closed loop, it does not fall back towards the +0.28 EUR m⁻² and 0.15 survival of the library without cross terms, which is what a bilinear detour predicts. It scores +2.11 EUR m⁻² under the ensemble estimator and +2.28 under STLSQ, with survival 0.40 (8 of 20 replicates, Wilson interval [0.22, 0.61]; exact McNemar test against the full library $p = 0.55$): significantly above the middle library (mean paired difference +1.83, 62 of 80 pairs, $p_{\text{Holm}} = 4.4 \times 10^{-6}$, Holm correction over the four contrasts of this block) and at most slightly below the full library (median -0.18, $p_{\text{Holm}} = 6.4 \times 10^{-3}$ under the ensemble estimator; not separable under STLSQ, $p = 0.089$). The small significant cost of the deletion in one estimator arm suggests the bilinear term shares the threshold's pressure with the direct term rather than replacing it. Which feature, or which correlation pattern, makes the threshold sever the direct term in the library without cross terms is therefore open. Across four libraries the margin nevertheless continues to track survival—0.15 $\rightarrow$ +0.28, 0.40 $\rightarrow$ +2.11, 0.55 $\rightarrow$ +2.75 and +4.32 EUR m⁻²—while neither conditioning, one-step accuracy nor library size orders them. The cross-term block already present in the mechanism experiment could not have substituted for this wave: that block edits coefficients on a $\lambda = 10^{-6}$ fit in which the direct boiler term survives in all 20 replicates, so the pathway is never severed by sparsification there, and none of its contrasts reaches significance under either objective.

4.2. A cheap open-loop rule for selecting a control-oriented surrogate

Two things follow for practice, and they are different in kind. The first is a screening rule: score candidates open-loop on (i) free-running multi-step rollout error at a horizon no shorter than the one the model will be used at and (ii) the fraction of rollouts that diverge—both computed on the identification data in seconds, before any closed-loop season is simulated—and keep one-step error out of the gate. The condition number belongs in the report as a diagnostic but not in the ranking: it predicts (i) and (ii) accurately and closed-loop margin not at all (Section 4.1). Ours scored 96 steps (24 h) against a 20-step (5 h) control horizon—a stress test rather than a matched one, so the screening horizon has to be reported with the score. In our ladder the raw configuration dominates the frozen physics-informed recipe on all three pre-registered axes at once (Section 3.1), and the two controllers built on that library then take the top two places in closed loop. The frozen recipe leads on exactly one metric, one-step accuracy (rank 8 of 72 against rank 10), which was not a gate. The screen is close to free: the whole ladder cell costs a few seconds per fit, against a simulated season per candidate for a closed-loop evaluation.

The strength of that claim must be stated at the resolution the three libraries allow. Over the series, the screen identifies the closed-loop winner and one-step accuracy ranks that same winner last; but neither open-loop criterion reproduces the full closed-loop ordering, because rollout error places full physics third where the closed loop places it second. What is established is the choice of the top candidate and the rejection of one-step accuracy as a selection criterion—not a rank correlation across the series.

The second thing that follows is not a ranking but a check, and it is the one the new mechanism makes necessary. After sparsification, read the fitted coefficient matrix and confirm that every actuator the controller is allowed to move still has a surviving path to the states it is meant to move. The check costs nothing—it inspects numbers already computed—and it is the variable the closed loop turned out to be sensitive to, whereas a well-conditioned regression is not a guarantee that the check passes and an ill-conditioned one is not a guarantee that it fails. When the check does fail, the available responses are a lower

threshold, a different basis, or a constraint protecting the actuator columns; we tested only the first, by hand, on one coefficient, and it bought back a median of $+0.21 \text{ EUR m}^{-2}$.

Two qualifications. The ladder's rollout value is a *comparator*, not a specification: along the oracle controller's own closed-loop trajectory the same frozen recipe predicts indoor temperature to $2.62 \text{ }^{\circ}\text{C}$ at 24 h, four times better than the $10.50 \text{ }^{\circ}\text{C}$ it scores on PRBS-excited identification data. The held-out block supports the ordering but licenses a different claim: the ordering is identical in-sample and held-out, yet the dominant factor is the *evaluation* year. The physics library without cross terms scores a median rollout error of $16.9 \text{ }^{\circ}\text{C}$ evaluated on 2018 and $2.7 \text{ }^{\circ}\text{C}$ on 2019 whatever year it was fitted on, and in one of the four fit→evaluate directions it narrowly beats the raw library (2.687 against 2.722). What is defensible is that the raw library is stable regardless of evaluation year while the physics libraries are year-dependent.

The third piece of guidance is therefore negative: do not treat a basis function's physical motivation as evidence that it belongs in the library. Every feature added here is an algebraic function of columns already present, and the two groups in which they were added both cost multi-step stability—the three non-cross features take rollout error from 3.31 to $7.11 \text{ }^{\circ}\text{C}$, the four cross terms take it on to 30.84 —while neither group buys back a closed-loop margin above the raw library's. Nor does the *count* of added features order the damage: adding three is worse in closed loop than adding seven (Section 4.1), so a practitioner cannot reason from “I added only a few, physically sensible terms” to “the cost is small”. What we cannot say is which individual feature is responsible for anything: they entered in two groups of three and four, no feature was added or removed alone, and the one single-feature deletion that would matter most is the untested experiment named above. Motivation earns a candidate a place in the ladder, and nothing more.

4.3. Sparsity is exonerated

The winner is the sparser model on both counts—11 candidate features against 14, and 19.9 retained terms against 28.25—so sparsity cannot be what the physics-informed variants lost. Nor is more sparsity better within a library: over the sparsity sweep of Table 10 margin falls as terms are removed, and violation steps rise with it, from about 3850 to 9090 between the endpoints. The decline is not monotone across the thirteen levels, and the between-replicate SD reaches 4.14, larger than several of the steps, so only the endpoints carry the comparison. The three-library series settles the question in the other direction as well: across the ensemble triple, in which the library is the only thing that changes, retained terms run $20.2 \rightarrow 28.25 \rightarrow 37.2$ while closed-loop margin runs $+4.32 \rightarrow +0.28 \rightarrow +2.75 \text{ EUR m}^{-2}$. The least sparse model of the three is not the worst; the middle one is. Term count orders performance in neither direction; the basis does. The two raw-library variants—bagged ensemble and plain sequential thresholded least squares—cannot be separated in closed loop (median $\Delta = -0.01$, mean $+0.48 \text{ EUR m}^{-2}$, 33/80 wins, $p = 0.60$), which locates the effect in the library rather than in the estimator. This is a failure to reject, not a demonstration of equivalence: the mean gap is not negligible, only too disperse to establish at $n = 80$.

4.4. Sensitivity to baseline tuning and economic weighting

Two benchmarking safeguards materially changed the comparison (Table 15, Figure 5): tuning the agronomic reference and aligning MPC resource weights with the prices used for evaluation. Figure 5 renders the two changes as cascades: panel a subtracts the gain from tuning the baseline from the raw-library advantage, and panel b subtracts the effect of price alignment from the raw-minus-physics gap.

Searching the heuristic's four economically relevant setpoints under the budget assigned to the reinforcement-learning agents—16 random trials, selection on the training years only—is worth $+3.49 \text{ EUR m}^{-2}$ and moves the heuristic from last-but-three to

seventh of fifteen. The apparent advantage of the raw-library controller over it falls from +5.54 to +2.06 EUR m⁻². Thus, the unsearched baseline would overstate the advantage by a factor of 2.7. Only 2 of the 16 random draws beat the stock setpoints: the stock configuration sat at the 88th percentile of its own search space: it was a credible agronomic baseline even before the search. Tuning also moved the heuristic *onto* the two-axis front, where untuned it had been dominated. Tuning budgets should be reported in units comparable to those granted to the learned controllers [45].

The default MPC stage cost weighted boiler, lamp and CO₂ in the ratio 1:0.5:0.1, while the marginal costs used by the evaluation criterion are 1:2.97:0.46. Re-weighting the resource term to those ratios, with its total at $u = (1,1,1)$ held fixed, improved *both* axes for all eight MPC controllers, but unevenly: the five physics-informed variants gained +1.54 ... + 2.74 EUR m⁻² and the neural-network MPC +1.57, against +0.21 ... + 0.25 for the two raw ones. Taking the best physics-informed variant, the raw-minus-physics gap fell from +3.66 to +1.23 EUR m⁻². The tuned heuristic also left the two-axis front, yielding to a physics-informed variant it had beaten under the default weights. Why price alignment benefits one library more than another was not isolated. The practical implication is nevertheless clear: a greenhouse-controller benchmark should use a tuned agronomic reference and site-relevant energy and CO₂ prices before attributing an economic difference to the surrogate model class.

4.5. Relation to prior work

The finding instantiates the identification-for-control programme—model quality cannot be judged independently of the control task the model serves [18–20,25,46]—and the objective-mismatch literature of model-based reinforcement learning, where prediction likelihood is a poor proxy for control performance [21–24]. Our contribution there is constructive rather than diagnostic: an open-loop screen costing seconds per candidate identifies the closed-loop winner, whereas the standard screen ranks that winner *last*. The screen needs only free-running rollouts, so it transfers in principle to any model class; but we exercised it on sparse-regression libraries, and over the three of them it selects the winner without reproducing the ordering of the two losers (Section 4.2), so the correspondence is established as a choice of top candidate, not as a rank correlation and not across model classes. Section 4.4 adds a second, less-studied mismatch: stage-cost weights against the prices in the criterion, inside an otherwise correctly specified problem. Collinearity in the library is what reweighted and balance-guided sparse regression is designed to mitigate [28–29,47], and our open-loop measurements agree with that literature exactly: κ orders multi-step stability without exception. What we do *not* find is the further step from conditioning to closed-loop economics, where the ordering breaks (Section 4.1). The consequence is not that library design matters less but that it has to be scored on what the controller actually uses—a point on which library optimisation scored by recursive long-term prediction [17] converges from the opposite direction. Bagged ensemble [13] changed neither outcome: it is statistically indistinguishable from plain thresholded least squares on the raw library ($p = 0.60$) and on the full physics library ($p = 0.057$). Greenhouse comparisons of MPC and reinforcement learning have generally fixed the model and varied the controller [6,14–16]; our results show that the surrogate library can be comparably consequential.

The study also belongs to a developing simulation-based literature in *Agronomy*. Xu et al. used a mechanistic greenhouse–crop model to compare open- and closed-loop optimal control and reported simulated yield and profit [7]; Wen et al. combined an experimentally fitted photosynthesis surrogate with multi-objective regulation of light and CO₂ [8]; and Ećim-urić et al. coupled transient building simulation with an ANN after comparing modelled and measured greenhouse temperatures [9]. The present contribution

differs by holding the mechanistic greenhouse fixed and treating surrogate selection itself as the experimental factor. It also provides a lower level of empirical evidence than studies that operate controllers on a crop, such as the eight-week reinforcement-learning irrigation evaluation of Padilla-Nates et al. [48]. Accordingly, our results support model and controller design choices for subsequent testing; they do not establish agronomic efficacy in a physical greenhouse.

4.6. Limitations

- **There is no physical-greenhouse validation.** Every climate trajectory, crop response, resource use and margin is generated by GreenLight under ERA5-derived weather. The published evaluation of GreenLight supports its use as a common testbed, but does not validate the controllers developed here. The comparison excludes unmodelled sensor bias, actuator wear and delay, crop heterogeneity, pests and disease, and operational intervention by growers. Absolute EUR m^{-2} values should therefore be read as outputs of the virtual experiment; only comparative behaviour within the stated model and price assumptions is established. Hardware-in-the-loop testing followed by a greenhouse pilot with independently measured climate, energy use and marketable yield is required before deployment.
- **The advantage is not Pareto-dominant.** Five controllers hold the front; PPO incurs 1330 violation steps against the raw library's 4130, at 3.86 EUR m^{-2} less margin. Superiority is conditional on how a grower weights margin against corridor excursions.
- **The MPC objective is necessarily a proxy.** The surrogate carries no crop state, so revenue cannot be predicted and terminal biomass cannot be priced—structural to the model class, and it blocks the natural terminal-cost formulation.
- **The tuned-heuristic comparison is not a paired test.** The heuristic is deterministic and ran once per season, so its value is a constant reused across replicates and carries no sampling uncertainty: the $+2.06$ above is a one-sample signed-rank against that constant. Genuinely paired it collapses to one replicate's four seasons ($+2.55 \text{ EUR m}^{-2}$, $n = 4$), where the minimum attainable p is 0.125 and nothing can be significant.
- **Significance is not magnitude.** Two physics-informed variants differ by $0.0066 \text{ EUR m}^{-2}$ over four seasons yet separate at $p = 0.012$, so listing both as front points is an artefact. Effect sizes accompany every p -value, and Holm corrections are over a declared family of 15 controllers—another declaration moves them. This is not hypothetical: adding the two full-physics controllers enlarged the family from 13 to 15 and moved every corrected level in the main comparison, the raw-versus-physics contrast from 7.7×10^{-4} to 4.6×10^{-4} among them, without any controller crossing 0.05 .
- **Coverage is uneven.** The neural-network MPC has 10 replicates and 40 runs against 20 and 80 elsewhere, 14 of them terminating early. The oracle has no 2022 season—all 20 runs exhausted the solver-failure budget—but that is an artefact: at a thousandfold budget the season completes at -1.12 and -0.56 EUR m^{-2} , so the reported -2.65 is a three-season mean and the imputed four-season value is ≈ -2.20 . One front member owes part of its position to truncation: over complete runs its violation margin narrows from 71 to 26 steps.
- **The mechanism rests on a three-point library comparison, which is few.** A claim about what does and does not order closed-loop performance is being carried by three libraries, and the third point exists on one optimiser only: the ensemble triple is complete, whereas at threshold 0.05 under sequential thresholded least squares only the raw and full-physics libraries were run ($+3.83$ and $+2.48 \text{ EUR m}^{-2}$), so the raw-over-physics direction replicates but the non-monotonicity itself does not exist as a three-point series on that axis. Boiler-term survival is *measured, not manipulated*, at library level: the libraries differ in the basis, and survival is a downstream property of the resulting fits. The only randomised manipulation anywhere in this argument is the single-coefficient ablation, whose median effect is $+0.21 \text{ EUR m}^{-2}$ —so the mechanism is supported by one

controlled contrast of modest size plus a library pattern that is measured rather than manipulated. A fourth library has already been added—the term-deletion probe below—and did not disturb the pattern, but the pattern remains observational at library level.

- **What sets boiler-term survival is unresolved.** The candidate structural explanation—that the bilinear $T_{in}u_{Boil}$ term is what rescues the full physics library—has been tested by deleting exactly that feature, same optimiser and threshold, four seasons $\times$ 20 replicates in closed loop, and is rejected (Section 4.1): the 17-feature library keeps the full library’s conditioning, open-loop instability and closed-loop level, and its survival (0.40) comes nowhere near the middle library’s 0.15. The two observational figures once offered for the detour were non-significant at 9–11 and 3–17 replicates and now stand only as absence of contradiction. The non-monotonicity is an established fact without an established cause.
- **The mechanism block is single-season.** All 800 mechanism runs are for 2020, so neither the λ sweep nor the ablation generalises across seasons. The knock-in and cross-term blocks rest on different baselines (boiler-term survival 0.15 against 1.00) and are not mutually comparable; the split by term survival is observational and confounded with recipe, and only the knock-in contrast is controlled. The library comparison, by contrast, covers all four test seasons, including the term-deletion probe.
- **The held-out check rests on two training years.** With 2018 and 2019 alone, year-dependence cannot be separated from chance, and the evaluation year dominates (Section 4.2).
- **The design and sensitivity blocks use the default objective.** The support harness hard-codes the default stage cost, so those sensitivities do not test the price-aligned formulation. Under the default objective, perturbing identified coefficients by $\pm 15\%$ and $\pm 20\%$ degrades margin significantly (9 of 10 and 10 of 10 replicates worse, span of means 12.2 EUR m^{-2}), but as a growing minority of catastrophic runs, not a uniform decline—the median at $\pm 20\%$ is only -0.31 . This finer grid stops at $\pm 20\%$, so the non-monotonicity the coarser canonical grid shows at $\pm 30\%$ is unreproduced rather than refuted.
- **The horizon was never tuned for any MPC.** Every MPC improves at horizon 8 over horizon 20, by $+0.29$ for the raw library and $+1.3 \dots + 2.1$ for the physics-informed variants—consistent with rollout error compounding inside the prediction horizon, which is the open-loop half of the argument and is unaffected by Section 4.1; and an asymmetry running against the MPCs, which received no hyperparameter search while the reinforcement learning agents and the heuristic received 16 trials each.
- **Reproducibility is within-environment only, and cross-environment reproducibility was never measured.** The pipeline reproduces nine digests bit-for-bit on one machine. Whether closed-loop margins survive a change of machine is *unknown*: every wave reported here ran in one environment, and the manifests do not even record which one (Section 2.9). What is measurable in the tree is that closed-loop margins move under changes far smaller than a change of platform—the two harnesses score the *deterministic* heuristic differently in all four seasons at identical configuration hash, by up to 0.038 EUR m^{-2} and monotonically more with each season. A controller with no stochastic element reproducing itself only to two decimal places is the conservative reading of this limitation, and it is enough to warrant the rule adopted: whole waves are regenerated in one environment and never mixed cell-by-cell. The configuration hash does not discriminate between objective variants either, so the results tree and block, not the hash, are the identifying key.
- **Negative results stand as such.** The out-of-distribution guard does not pay for itself, and the sign of its effect depends on which statistic is taken — mean against median, all pairs against completed seasons (Section 3.7). It truncates a left tail rather than raising the centre, costs violations, ends 30 % of its seasons early, and rests on a detector barely better than chance; its activation count was never recorded, so the natural explanatory variable is missing. The extended Kalman filter observer is worse than doing nothing (-2.95 against -1.58 EUR m^{-2}), 39 % of its seasons terminating early.

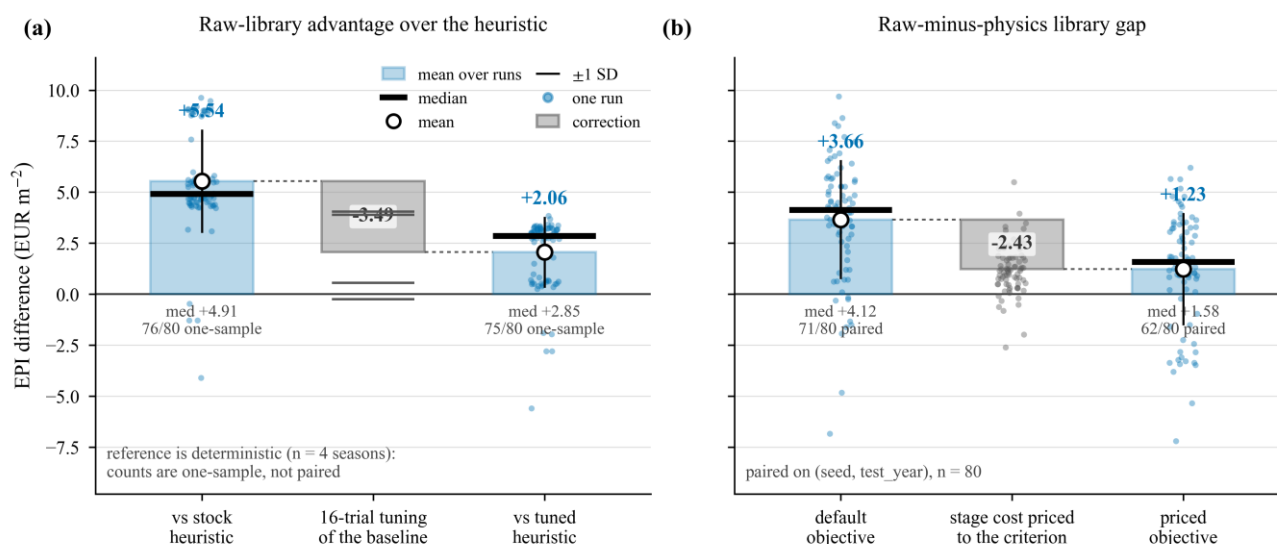
- **Additional scope limits.** One location, one planting date, one crop model and one weather source are represented. The ranking is not price-invariant: two different controllers win across a nine-point grid of fruit and energy prices—though that grid re-scores rather than re-optimises and contains neither raw-library controller, so it does not test the ranking reported here. No recursive feasibility or stability guarantee is claimed or proved [4].

Table 14. The selection reversal, measured on the held-out identification block. Rollout RMSE is the median over replicates at the 96-step (24 h) horizon for indoor temperature, with the interquartile range beside it; κ is the condition number of the regression matrix; “terms” counts non-zero coefficients after thresholding; $n = 40$ per cell (10 replicates $\times$ 2 sparse estimators $\times$ 2 fit years). κ and the term count are properties of the fit and so repeat across regimes. The ordering by one-step error is the exact inverse of the ordering by every multi-step criterion, in both regimes—but only over the two sparse estimators used here; pooled over all four estimators of the full ladder the one-step ordering is no longer strict. The medians overstate the separation: both physics libraries are strongly dispersed, and the lower quartile of the no-cross library falls to the raw library’s level, which Section 4.2 attributes to the evaluation year. The table is an *open-loop* measurement throughout: κ orders every column here, and it does not order closed-loop margin, where the series is non-monotone (Section 4.1). The column is reported as a diagnostic and must not be read as a performance ranking.

Feature library	Features	κ	Terms	One-step RMSE (°C)	Rollout RMSE, median [IQR] (°C)	Diverged (fraction)
<i>In-sample</i>						
Raw (physics-free)	11	8.2	20.6	1.84	2.62 [2.53, 2.80]	0.000
Physics, no cross terms	14	25.4	29.7	1.69	12.14 [2.88, 14.10]	0.021
Physics, full	18	56.2	36.6	1.64	21.88 [18.06, 25.58]	0.078
<i>Held-out year</i>						
Raw (physics-free)	11	8.2	20.6	1.90	2.72 [2.63, 2.84]	0.000
Physics, no cross terms	14	25.4	29.7	1.82	15.42 [2.69, 21.13]	0.046
Physics, full	18	56.2	36.6	1.77	21.57 [15.56, 31.74]	0.086

Table 15. Sensitivity of the headline comparison to two benchmarking safeguards. EPI is the simulated seasonal margin returned by the virtual greenhouse.

<i>Safeguard 1: tuned baseline</i>		<i>Safeguard 2: price-aligned stage cost</i>
Risk controlled	comparison with an otherwise credible but unsearched agronomic heuristic	actuator weights of 1: 0.5: 0.1 (boiler: lamp: CO ₂) differ from marginal-cost ratios of 1: 2.97: 0.46
Implementation	16 random trials, the budget the reinforcement-learning agents received; selection on training years only. The MPCs received no search at all, an asymmetry running against them	energy term re-weighted to the measured ratios, its total at $u = (1,1,1)$ held fixed
Direct effect	heuristic $-1.23 \rightarrow +2.26$ EUR m ⁻² (+3.49)	both axes improve for all eight MPC controllers, every contrast Holm-significant, but by $+1.54 \dots + 2.74$ for the physics-informed libraries against $+0.21 \dots + 0.25$ for the raw ones
Effect on the comparison	raw-library advantage over the heuristic $+5.54 \rightarrow +2.06$ EUR m ⁻² ($2.7 \times$ smaller)	raw-minus-physics gap $+3.66 \rightarrow +1.23$ EUR m ⁻² (two-thirds removed)
Also changed	the heuristic moves onto the two-axis front, which it had not held untuned	it leaves that front again, dominated on both axes; and the boiler-coefficient ablation falls from median $+3.05$ to $+0.21$ EUR m ⁻²



Positive = the raw library is ahead. Panel (a): both ends from n2_tune/tune_rb_n2.csv so the waterfall closes exactly. Panel (b): n7/main_n7.csv and final/main.csv (default) against the priced pool.

Figure 5. Reduction of the headline contrast by two benchmarking safeguards. (a) Tuning the agronomic reference. (b) Aligning MPC stage-cost weights with evaluation prices. Bars show the contrast before adjustment, the safeguard effect and the residual contrast.

5. Conclusions

Within a GreenLight-based in silico greenhouse testbed, we asked which open-loop criterion should select a data-driven surrogate for economic MPC and which model property best explains the closed-loop result. Multi-step stability selects the leading candidate, while survival of the actuator term through the sparsity threshold—rather than conditioning alone—best tracks the non-monotone economic outcome (Figure 6).

- One-step accuracy and multi-step stability rank the candidate models in opposite orders.** In the first-order, undenoised block of a 72-configuration ladder, under two sparse estimators, physically motivated features lowered one-step RMSE from 1.86 to 1.68 °C but raised the median 24 h rollout RMSE from 2.67 to 24.27 °C. The reversal is confined to that block—pooled over all 72 labels the raw library predicts better one step ahead too—but it reproduces on a held-out year [17,19–21,24].
- The library selected by multi-step stability also gives the best simulated closed-loop economics under the benchmarking safeguards.** Across fifteen controllers on four test seasons the raw-library ensemble surrogate [13] led with +4.32 EUR m⁻², against +3.09 for the best physics-informed variant (Holm-corrected $p = 4.6 \times 10^{-4}$, family of 15) and +2.26 for the agronomic heuristic, re-tuned under the reinforcement-learning agents' own 16-trial budget ($\Delta = +2.06$, 75 of 80 runs against a deterministic reference) [33–34]. Tuning was worth +3.49, and aligning the actuator weights with the evaluation prices cut the raw-minus-physics gap from +3.66 to +1.23. The direction is retained, but its magnitude is sensitive to baseline tuning and economic weighting.
- Physically motivated features can reduce performance non-monotonically, and conditioning does not order the economics.** Changing the feature library alone—same estimator, threshold and seeds—the condition number rises monotonically, 8.2 → 24.5 → 53.4, the fits grow denser rather than sparser (20.1 → 37.0 terms), and open-loop rollout error rises with them, 2.67 → 10.58 → 24.27 °C. Closed-loop margin does not: +4.32 → +0.28 → +2.75 EUR m⁻², the worst-conditioned library beating the intermediate one almost tenfold, with the same shape under the other estimator. Conditioning predicts open-loop stability cleanly but does not order closed-loop economics [28–29,47].

4. **The closed-loop outcome follows whether the actuator pathway survives thresholding.** The direct boiler co-efficient survives in 55, 15 and 55 % of fits of the three nested libraries — the ordering of the margins, which conditioning does not reproduce — and in 40 % of fits of the 17-feature deletion library, whose margin (+2.11 EUR m⁻²) lands between, again in survival order. Reinstating it in the recipe that loses it is significant but small and right-skewed: median +0.21 EUR m⁻², mean +1.92, 16 of 20 replicates, $p = 5.9 \times 10^{-4}$, one season. Stratifying that recipe's own fits by survival agrees, not significantly ($p = 0.053$). A candidate bilinear temperature–boiler detour was tested by deleting that single feature and is rejected; why the middle library loses the term is open, that survival gates the outcome is not.
5. **The advantage is neither Pareto-dominant nor unconditional.** The margin–violation front holds five of the fifteen controllers, and PPO [30] incurs 1330 violation steps against the raw library's 4130 at 3.86 EUR m⁻² less margin — though two of those points are one controller under two thresholds and a third rests partly on seven truncated runs. The surrogate has no crop state, so terminal biomass is unpriced; the out-of-distribution guard is a negative result (paired mean −0.69 EUR m⁻², +430 violation steps); a $\pm 15\%$ coefficient perturbation costs 7.97 on the mean against 3.35 on the median ($p = 4.3 \times 10^{-3}$); and all of it is simulation, one location, one planting date, one computing environment.

These findings establish a control-oriented model-selection result within the stated virtual greenhouse. They do not demonstrate realised marketable yield, energy savings, constraint safety or profitability in a physical facility. Before operational use, the selected surrogate and controller require calibration against measured greenhouse trajectories, hardware-in-the-loop testing and a crop-scale pilot that independently records climate, actuator use, energy consumption and marketable yield.

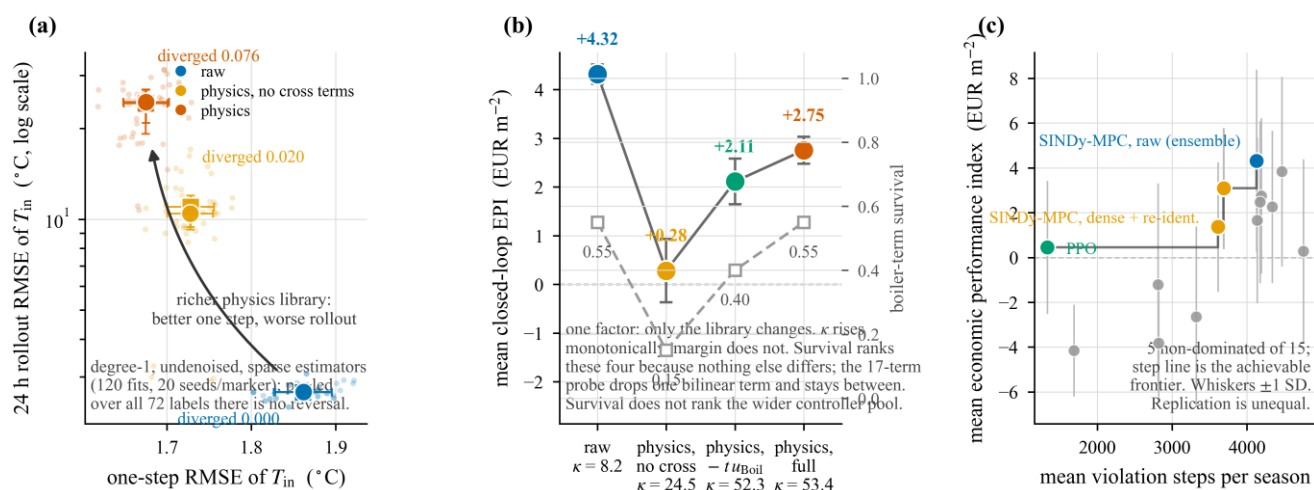
Table 16 lists every headline quantity above together with the result file it is read from, so that each can be checked at source.

Table 16. Headline quantities of the study and the result files they are read from. All paths are relative to own-article/regen/results/. EPI is the seasonal economic performance indicator (margin) returned by the simulator; n counts runs after deduplication on (method, seed, test year). Where three values are given they are the three feature libraries in the order raw / physics-no-cross / physics.

Quantity	Value	Source file
<i>Identification ladder</i> (degree 1, no denoising, sparse estimators, $n = 40$ per library)		
One-step RMSE, T_{in} (°C)	1.86 / 1.73 / 1.68	ladder_rerun/ ladder_rerun*.csv
Median 24 h rollout RMSE (°C)	2.67 / 10.58 / 24.27	as above
Divergence fraction	0.000 / 0.020 / 0.076	as above
Condition number κ	8.21 / 24.52 / 53.43	as above
Non-zero terms	20.1 / 28.2 / 37.0	as above
<i>One-factor library change</i> (ensemble, threshold 0.05, priced objective, four seasons, $n = 80$ per library, zero truncated runs). Only the feature library differs between these three controllers.		
Closed-loop EPI (EUR m ⁻²)	+4.32 / +0.28 / +2.75	priced_main/*.csv, phys_lib/ main_physlib.csv
Boiler-term survival ^b	0.55 / 0.15 / 0.55	as above
The same change under STLSQ (EUR m ⁻²)	+3.83 / — / +2.48	as above
<i>Four test seasons 2020–2023</i> , fifteen controllers ($n = 80$ unless stated). MPC rows use the priced stage cost; PPO, SAC, the heuristic and the full-model planner have no MPC stage cost and are quoted from the canonical wave.		
Raw-library ensemble MPC (EUR m ⁻² ; violation steps)	+4.32 $\pm$ 4.08; 4130	priced_main/*.csv
Best physics-informed MPC (EUR m ⁻²)	+3.09 $\pm$ 2.71	priced_main/*.csv
Raw ensemble vs. that variant	$p_{Holm} = 4.6 \times 10^{-4}$	recomputed over the family of 15

Quantity	Value	Source file
Tuned heuristic (EUR m ⁻²)	+2.26 ($n = 4$) ^a	n2_tune/tune_rb_n2.csv
Stock heuristic (EUR m ⁻²)	−1.23 ($n = 4$) ^a	n2_tune/tune_rb_n2.csv
PPO (EUR m ⁻² ; violation steps)	+0.45 ± 2.97; 1330	final/main.csv
<i>Unseen seasons 2014–2017, default objective ($n = 32$, 8 replicates, four controllers only)</i>		
Raw-library ensemble MPC (EUR m ⁻²)	+6.42 ± 0.95	n5_years/main_n5.csv
Untuned heuristic (EUR m ⁻²)	−0.33	n5_years/main_n5.csv
<i>Benchmarking safeguards, mechanism and robustness</i>		
Gain from tuning the heuristic (EUR m ⁻²)	+3.49	n2_tune/tune_rb_n2.csv
Raw-minus-physics gap, default → priced (EUR m ⁻²)	+3.66 → +1.23	priced_main/*.csv, priced_dagger/*.csv, final/main.csv, n7/main_n7.csv
Boiler-coefficient knock-in, priced: median / mean	+0.21 / +1.92 (16 of 20, $p = 5.9 \times 10^{-4}$)	priced_mech/ mechanism_pricedMech*.csv
±15 % coefficient perturbation, priced: mean / median cost (EUR m ⁻²)	−7.97 / −3.35 ($p = 4.3 \times 10^{-3}$)	design_priced_real/ design_designPriced.csv

^a The heuristic is deterministic and was run once per season, so $n = 4$ and no run-to-run dispersion exists; the dispersion of its seasonal values must not be read as an error bar, and any signed-rank test against it is a one-sample test against a constant. Both heuristic rows are quoted from the same harness, n2_tune, so that the tuning gain is internally consistent; the canonical wave returns −1.21 rather than −1.23 for the identical deterministic controller, a 0.02 EUR m⁻² discrepancy discussed with the reproducibility limits. The two values must not be mixed within one table. ^b Fraction of identification replicates in which the direct boiler-to-temperature coefficient is non-zero after thresholding. Survival is a property of the fit rather than of the run, so its effective n is 20 seeds, not 80 runs.



Graphical Abstract. (a) Open-loop selection reversal. (b) Non-monotone closed-loop economics and boiler-term survival. (c) The five-member margin-violation Pareto front.

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authors have read and agreed to the published version of the manuscript.” Initials must match the author block above, and every listed author must appear in at least one role.

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Institutional Review Board Statement: Not applicable. This study is a computational study of a greenhouse climate simulator and involves neither humans nor animals.

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Data Availability Statement: All quantities reported in this article are computed from the per-run result tables of the study’s regeneration tree (own-article/regen/results/ in the project repository), which contains one row per (controller, seed, test season) together with the seasonal margin, constraint-violation counts, solver diagnostics and the identified sparse coefficients. Every wave reported here was produced under a single frozen configuration whose hash, 637c6b535a9e, is written into every result row and into the regen_manifest.json of each wave alongside the git commit of the code that generated it; the manifest also records the seeds, test years, horizon, solver-failure budget, season length and the four identification recipes, so a wave can be identified without reference to this text. The experiments are regenerated with the single driver in own-article/regen/: each of the eight experiment blocks is run as `python run_regen.py --experiment <block> --seeds <list> --out <dir>`, after which `python run_regen.py --merge --out <dir>` and `python make_tables.py --out <dir>` rebuild every derived table together with NUMBERS.md, a claim-to-value-to-source map that names the file and column behind each reported quantity. `python verify_regen.py` then applies the acceptance gates—mixed configuration hash, an incomplete or duplicated run grid, truncated seasons, non-uniform horizon or solver budget, a missing sparse coefficient column, an unresolved sparsity region or an incomplete ablation block—and exits non-zero if any fails. Determinism is checked separately by `python repro.py --selftest`, which executes the pipeline twice under identical seeds and compares SHA-256 digests of the training states and actions, the ensemble and STLSQ coefficient matrices, the neural-network weights, and the closed-loop trajectory and margin; all seven digests agree on the pinned stack, and all nine with the two reinforcement-learning policies included. Reproduction additionally requires the GreenLight tomato greenhouse model as packaged in `gl_gym` 0.3.1 [1,16] and the ERA5-derived weather described in Section 2.1, retrieved from the Open-Meteo historical archive API [36–37]. As Section 2.9 states, bit-level reproduction is established within one computing environment; the cross-environment case was not measured, and closed-loop margins should not be expected to match to the last decimal on a different stack. [[REQUIRED BEFORE SUBMISSION: a public, citable location for this tree — a repository URL and an archived release with a DOI (Zenodo, figshare or equivalent). MDPI requires a link or an explicit statement of restriction; a path inside a private repository is not sufficient.]]

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Conflicts of Interest: [[REQUIRED — NOT SUPPLIED.]] Every author must declare. Use “The authors declare no conflicts of interest.” only if that is true of all of them; otherwise describe the interest and state the funders’ role, or lack of role, in the design of the study, in the collection, analyses or interpretation of data, in the writing of the manuscript, and in the decision to publish the results.

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